CM-SIP — Study Guide

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CM-SIP — Study Guide

Capital Markets – Specified Investment Products (SIP): Derivatives and Collective Investment Schemes

A concise guide to the CM-SIP examination: derivatives, structured products and collective investment schemes. It condenses the official study guide into what the examination tests, section by section.

- **Exam:** 80 multiple-choice/multiple-response questions in 2 hours 30 minutes, plus 6 case studies containing 18 questions (3 per case study); **70% to pass**.
- **Currency of the material:** the official guide states its regulatory information as current at 30 September 2023. Check the latest MAS, SFA, SGX and product terms before relying on a rule or specification.
- This is a study aid, not a replacement for the prospectus, Product Highlights Sheet (PHS), term sheet, risk disclosure statement or current law. If you will conduct regulated activity, you must also meet the relevant MAS representative-notification and competency requirements.

How to use this guide

1. Learn the product mechanism before memorising the payoff.
2. For every product, ask: what is the underlying, what is the investor's position, what triggers payment/conversion/termination, what can be lost, and who is the counterparty?
3. Practise the formulas in Appendix D and the application cases in Chapter 12.
4. Use `CMSIP_QUESTIONS.md` only after reading the relevant section; its answers explain the reasoning rather than merely naming a letter.

Examination orientation

- The examination tests knowledge, understanding and application of SIPs.
- Format: 80 multiple-choice/multiple-response questions in 2 hours 30 minutes, plus 6 case studies containing 18 questions (3 per case study); the passing mark is 70%.
- Unless a question says otherwise, use a **365-day** convention for Singapore Government Securities and a **360-day** convention for other products.
- A non-programmable financial calculator may be used and a formula sheet is provided.
- Scope: exchange-traded and OTC derivatives, structured products and collective investment schemes. Candidates who will conduct regulated activity must also satisfy the relevant MAS representative-notification and competency requirements.

Master mental model

Every derivative or structured product transfers one or more of these exposures:

Exposure	Typical instrument or feature	Main question
Direction	Futures, forwards, calls, puts, CFDs, warrants	Does the investor profit from a rise or fall?
Volatility	Straddles, strangles, barrier options, volatility ETFs	Does the investor need movement or stability?
Time	Option time value, calendars, rollovers, callable notes	How does time decay, maturity or early terminati
Carry/yield	Interest-rate futures, swaps, structured deposits/notes	What is paid or received while the position is ope
Credit	CDS, CLNs, issuer/SPV obligations	Who may fail to pay?
Liquidity	OTC customisation, lock-ups, exotic derivatives	Can the position be closed at a fair price?
Currency	FX forwards/options/swaps, translated warrants, DCIs	Which currency is invested, settled and exposed?
Leverage	Margin, futures, CFDs, DLCs, warrants, accumulators	Can a small move create a large loss or margin ca

Chapter 1 — Overview of Derivatives

A derivative is a contract whose value is linked to the value or volatility of something else. This chapter builds the market map — the underlying asset classes, the exchange and OTC venues, the regulatory response to 2008, and Singapore's position — that every later chapter assumes.

Learning Objectives

The candidate should be able to:

- Describe common underlying asset classes for derivatives and structured products.
- Describe the different types of derivatives and their uses.
- Describe the characteristics of exchange-traded and over-the-counter derivatives.
- Discuss the development of the derivatives market in Singapore and beyond.

1.1 Introduction

A derivative is a financial instrument whose value is linked to the value or volatility of an underlying reference or asset. That reference may be a particular security, an asset, an interest rate, a combination of securities, a basket, an index, or another derivative. Derivatives trade either on centralised, regulated exchanges or bilaterally between counterparties in the less regulated over-the-counter (OTC) market.

Futures, forwards, swaps and options are the basic types. Beyond them sits a wide range of more complex or **exotic** derivatives, built by combining similar and different types. Barrier options, knock-in and knock-out products, swaptions, structured notes and structured funds are examples. By varying the features — strike price, tenure, currency, conditions for execution — and combining different derivatives, an issuer creates a wider range of payoff outcomes and risk profiles aimed at investors holding different views.

The economic uses are four:

- **Hedging** — insuring against adverse price, rate, currency or credit movement.
- **Speculation** — taking a view on the future direction of the market.
- **Arbitrage** — profiting from perceived price differentials between two or more markets or assets.
- **Repackaging** — combining instruments to meet specific investment needs and to create desired synthetic positions.

Investors also trade and hedge on **volatility** — their expectation of the degree of price movement, whether prices stay range-bound, swing wildly, or move steadily in one direction. As volatility trading on indices has grown, leveraged exchange-traded notes (ETNs) and leveraged exchange-traded funds (ETFs) have been created to amplify the impact of rising or falling percentage volatility. Many leveraged ETFs and ETNs actively use futures and options to replicate or track the direction — or the inverse direction — of the underlying index, which makes such funds and notes effectively derivative instruments.

1.2 Underlying Asset Classes for Derivatives and Their Uses

Reduced to building blocks, most derivatives are referenced from or crafted out of five basic asset classes: **foreign exchange, money market instruments, fixed income, equities and commodities**. Indices, baskets, funds and portfolio mixes are extensions and representations of these blocks.

Underlying assets	Commonly traded securities / indic
Currency	Major currencies (USD, EUR, JPY,
Money market, short-term Treasury instruments and their interest rates	Government T-bills (1, 3, 6, 9, 12 m
Fixed income (debt)	Sovereign and corporate debt instru
Equity	Listed stocks; equity or stock indice
Commodities	Commodities; commodities-linked st

Derivatives and structured products allow investors to:

- **a.** Hedge or insure against unfavourable price movements, or potential falls in the value of assets or a portfolio already held — futures, forwards, options.
- **b.** Adopt a view on the future direction of the underlying without investing directly in it.
- **c.** Augment potential profits beyond what pure exposure to the underlying can deliver — options, warrants.
- **d.** Change the payoff profile (cash-flow sequence, size, tenure, currency, risks) in ways the underlying asset cannot offer — swaps, futures, forwards, options.
- **e.** Profit from a fall in the value of the underlying asset or index by taking short positions — options, futures.
- **f.** Secure the right, without the obligation, to decide when to buy or sell the underlying, using a small outlay — long options and warrants. A short call or put is an obligation, not a right, should the buyer choose to exercise.
- **g.** Arbitrage, or take advantage of price differentials between two or more markets.
- **h.** Change the nature of an asset or liability to meet needs that standardised instruments cannot meet.
- **i.** Create synthetic positions without incurring the cost of buying or selling the underlying.

Two footnotes matter for scope. A **debt warrant** resembles an equity or company warrant but attaches to a bond or other debt security, giving the holder the right to buy more bonds or debt securities from the same issuer at a stated price; only equity and company warrants are covered in Chapter 5. **ETFs and ETNs**

are not derivatives, but they actively use derivative instruments to enhance or hedge — leveraged and volatility ETFs, for example, use futures and options to replicate or track an index.

1.2.1 Money Market and Other Short Term Interest Rate Instruments

These are short-term debt instruments issued by governments, financial institutions and corporations. They are highly liquid and less risky. They account for **more than 95% of futures and options traded on global exchanges**, and, according to the Bank for International Settlements (BIS), consistently for more than **three-quarters of OTC derivatives trades between 2015 and 2019**.

Minimum denominations are relatively large and transaction sizes can be substantial. Maturities range from **1 day to 1 year**, often under **90 days**. The market is dominated by interbank lending and trading in Treasury bills, commercial paper, banker's acceptances, negotiable certificates of deposit and repurchase agreements. Institutional investors dominate.

Many instruments are benchmarked against interbank rates — **SIBOR, LIBOR, HIBOR, TIBOR, EURIBOR** — the most widely used historically being **LIBOR** (London Interbank Offered Rate). LIBOR is an average interbank lending rate calculated from submissions of actual borrowing rates contracted between banks, which makes it vulnerable to manipulation where submitting banks collude. Colluding to manipulate interbank rates and indices in order to engineer outcomes in the derivatives market is illegal, and traders and dealers have been prosecuted.

LIBOR was phased out and ceased to be used in **2021**, replaced for dollar-denominated derivatives and loans by the **Secured Overnight Financing Rate (SOFR)**.

1.2.2 Foreign Exchange

Cash investments in local or foreign currency are the lowest common denominator for all assets. FX derivatives form the **second largest** category of derivatives traded globally.

Currency forwards and futures are the most basic form, generating different payoffs depending on expected currency movement over time. Swaps and options change the underlying currency or interest payoff under conditional triggers, such as the exchange rate reaching a particular strike. Trading in currency forwards, futures, swaps and options is largely dominated by institutional investors.

In emerging markets with tightly controlled capital flows, inconvertible currency and low liquidity, hedging local-currency risk can be difficult. This is partly solved with a **non-deliverable forward (NDF)**, created as a proxy for the emerging-market currency. NDFs trade OTC in offshore financial centres and settle in a foreign currency other than the NDF currency itself.

1.2.3 Fixed Income Securities

These are borrowings by issuers — mainly governments, government-linked entities or public institutions, financial institutions and large multinational corporations, spanning investment-grade and junk bonds. The holder provides a loan of principal to the issuer and receives periodic interest, with principal returned at maturity. Differences in size, frequency and timing of cash flow produce a return commensurate with the risk taken and the opportunity cost incurred. Fixed-income derivatives — futures, swaps and options — offer alternative cash flows, payoff profiles and modified risk exposures relative to the underlying instrument.

The industry has innovated continuously to populate the investment frontier, and investment banks in the **1980s** popularised creative repackaging of debt for resale. Banks bought long-term fixed-income securities with higher yield and used them as the underlying asset base against which they repackaged and issued smaller denominations of marketable securities. Coupon payments of long-term Treasury bonds were **stripped** to form the asset base for issuing zero-coupon bonds. Portfolios of real-estate loans were bought and used to issue **mortgage-backed securities (MBS)**. Consumer and student loans became the asset base for **collateralised debt obligations (CDOs)**. Even without derivatives there was creative repackaging creating new payoff profiles; read broadly, various CDOs and stripped zero-coupon bonds are *derived* from other bonds and so are fixed-income derivatives.

By the late **1990s**, credit derivative instruments were actively promoted. Investment banks repackaged and modified the credit risk of fixed-income securities using credit insurance and **credit default swaps (CDS)** issued by insurance companies. They also bought select high-yield securities — junk bonds rated **CCC and below** — mixed them with better-rated securities, and added CDS and credit insurance to create a new cocktail for resale.

The credit quality of the underlying asset, and the credit management process, ultimately determines the risk and value of the repackaged product. The **2008 Global Financial Crisis** illustrated this dramatically: defaults on real-estate mortgage loans and CDOs triggered the collapse of **Fannie Mae and Freddie Mac**, the bailout and sale of several global institutions, forced capital restructuring across the global banking community, and a subsequent massive rise in regulatory control and cost.

1.2.4 Equities

Equities let investors participate in company profitability. When earnings prospects and growth are positive the share price rises over time; when they are negative it declines.

Equities are more volatile than other traditional asset classes. The ability of equity derivatives to take **short positions**, or grant a right to sell, is particularly useful in hedging the downside volatility of stocks.

Equity futures let investors buy (long) or sell (short) equities for future delivery at an agreed price. Long equity call options and warrants give the choice to participate in upside with a smaller outlay, augmenting potential profit while limiting loss to that outlay. Long equity put options allow investors to take short positions.

In equity-linked structures, investors can attempt to enhance returns by **selling put options** to collect premium from willing counterparties when the market is flat. This secures yields much higher than typical deposit instruments, but the higher yield carries the obligation to own underperforming securities should they fall to critical trigger levels. Chapters 7 and 8 examine these instruments.

> **Note.** Shorting put options is not a hedge against downside risk. A short put is a bullish or neutral position that loses as the underlying falls.

1.2.5 Commodities

Commodities are the original asset class that spurred the creation of derivatives. Unlike financial derivatives, a significant portion of commodity derivatives trading comes from regular hedging needs and opportunistic demand from agriculture, mining and manufacturing — real underlying business demand.

Supply and demand determine the long-run direction of each commodity, but short-term trading is influenced by other factors, notably the movement of the **USD**. Supply shocks make commodities volatile — the decisions of the **Organisation of the Petroleum Exporting Countries (OPEC)** on oil prices being the classic case — which makes the ability to take short positions valuable to a commodity investor.

By convention most commodity futures are denominated in **USD**. When the USD appreciates against other major currencies, commodity prices tend to move in the opposite direction, and vice versa. In extreme shocks to currencies or equity markets, **safe-haven commodities** — gold, silver and platinum — tend to rise. Futures and options on these have become increasingly popular both to hedge and to take advantage of the volatility.

1.2.6 Trading with Indices

Index-linked products give exposure to an asset class, market or security without buying its constituents. Futures, option contracts, ETFs and ETNs are pegged to equity indices, commodity indices and popular currency pairs.

Such indirect exposure lets investors:

- **a.** Adopt cost-effective investment exposure, passively over the long term.
- **b.** Hedge and protect profits, or minimise losses, on an existing outstanding position.
- **c.** Take speculative short-term exposure to a short-term opportunity.
- **d.** Augment exposure through leverage — typically **2–3×** a reference index — using ETFs, ETNs, options, futures and warrants on the index of a particular underlying asset class.
- **e.** Speculate on the potential volatility, or lack of volatility, of a security or an asset-class index.

Volatility itself became tradable. The **Chicago Board Options Exchange Volatility Index (CBOE VIX, "VIX")** measures the market's perceived volatility of the S&P 500 in either direction, derived from the price movement of call and put options. VIX options and futures trade on the CBOE daily. When volatility rises the VIX rises; when market volatility falls the VIX falls. VIX options and futures let investors trade volatility regardless of market direction.

Leveraged ETNs and ETFs on the VIX were introduced to widen access — ETNs such as **XIV, ZIV, VIIX, VIIZ, TVIX, TVIZ**, and ETFs such as **SVXY, UVXY** — allowing investors to buy or short volatility, trading the rise or fall in the percentage volatility of the S&P 500.

These structures are extremely risky. On **5 February 2018** the VIX spiked and more than doubled in a single day. The short-volatility ETF **SVXY** (ProShares) and **XIV** (Credit Suisse) each fell **more than 80% in a single day**, wiping out a few billion dollars overnight across just two instruments.

Trading in both was halted temporarily. There was no recovery for holders of the XIV ETN at the open, because a **termination event** — a price level — had been triggered and the issuing bank closed out the synthetic product to protect itself from further downside, forcing traders caught on the wrong side to realise losses. In subsequent weeks, under pressure to address the highly risky leveraged position its product created, ProShares cut its leveraging of the short and long volatility ETFs **SVXY and UVXY by half**.

1.3 Exchange Traded Derivatives

Exchange-traded derivatives are standardised products, mainly futures and options, traded on centralised, regulated exchanges such as **Singapore Exchange Derivatives Trading (SGX-DT)** or **ICE Futures Singapore (IFSG)** — as opposed to bilateral OTC markets. Each product's characteristics are fixed by standard convention: contract size, underlying asset, delivery or expiration dates, and delivery arrangements.

At expiry every contract is settled, either by physical delivery or by cash settlement. Once a trade is executed between a buyer and seller on the exchange, it is recorded with the exchange's **clearing house**. The contract is ultimately not between the original buyer and seller but between the holders at expiry and the clearing house, which guarantees settlement as originally intended using **pooled initial margins** from both sides of the trade.

Global open interest in exchange-traded derivatives, measured in USD trillions, is published by the BIS in its derivatives statistics; the series runs from 2007 through 2020 for futures and options.

1.4 Over-the-Counter Traded Derivatives

OTC derivatives trade directly between two parties without an exchange. Transactions are seldom terminated or assigned — transferred to a third party — before maturity unless both counterparties agree. Banks, other large financial institutions, fund managers and corporations are the main participants. Historically the two parties settle bilaterally.

Before the credit crisis that began in **2007**, the OTC derivatives market was largely unregulated. In **2018**, OTC derivatives trading was estimated at **more than five times** the value of derivatives traded on global exchanges.

Currency and interest-rate forwards and swaps make up the bulk of OTC derivatives. OTC currency forwards and swaps are driven largely by trades between international banks; corporates hedging underlying currency positions and some sophisticated investors also participate, but they are the minority.

Credit default swaps are the next largest product group. Until 2007, CDS grew in popularity as institutions used credit insurance to enhance the credit rating and pricing of bonds. This ”selling off risk” became a lucrative business: insurers and financial institutions that sold credit insurance generated good revenues as long as there was no major systemic downturn. When mortgage-backed securities began to collapse, CDS — profitable for buyers — became a disaster for sellers, who were obliged to pay out on the insurance they had provided against default.

Characteristic	Exchanges	OTC marke
Contract terms	Standardised	Non-standa
Counterparty structure	Clearing house acts as central counterparty: matches trades, eliminates credit risk	Bilateral ag
Collateral	Margin required	Credit risk
Transparency	Transparent	Less transp
Regulation	Regulated	Largely unr

1.4.1 Regulation of OTC Derivatives Trading

Post-mortem reviews of the 2008 crisis attributed much of the blame to the proliferation of products and to the lack of monitoring and regulation of OTC financial derivatives — especially credit derivatives, sold by

investment banks to institutional clients and cascading down to private and retail clients. Instruments that began as tools for risk management became instruments for speculative gain. Warren Buffett's 2002 warning is the most quoted: derivatives are "financial weapons of mass destruction, carrying dangers that, while now latent, are potentially lethal."

In **September 2009** the G20 leaders agreed to push for reform of the global financial system. The derivative commitments were:

- greater transparency in derivatives trading;
- moving more standardised OTC contracts onto exchanges or electronic trading platforms;
- clearing through central counterparties (CCPs);
- reporting to trade repositories; and
- higher capital requirements for non-centrally cleared contracts.

1.4.1.1 Europe In **2012** the **European Market Infrastructure Regulation (EMIR)** set the framework for regulating OTC derivatives in Europe. In line with the G20 commitments, EMIR required European financial institutions and their counterparties to measure, report, monitor and clear obligations, and to manage operational risk and counterparty credit risk on OTC derivative transactions not cleared through a clearing house.

1.4.1.2 United Kingdom In **2013** the UK passed the **Financial Services (Banking Reform) Act**, requiring major universal banks to separate and **ring-fence** retail and commercial banking from investment banking.

1.4.1.3 United States The **Dodd-Frank Wall Street Reform and Consumer Protection Act** was introduced in **2010**. It enhanced the role of the **Commodity Futures Trading Commission (CFTC)** to cover regulation of swap trades.

In **2013** the Federal Reserve Board announced that banking entities covered by **Section 619** of Dodd-Frank must fully conform their activities and investments to the statute and regulations by **21 July 2015**. Section 619 — the **"Volcker Rule"** — prohibits insured depository institutions and their affiliates from engaging in short-term proprietary trading of certain securities, derivatives and other financial instruments for the firm's own account, subject to exemptions including market making and risk-mitigating hedging. It also imposes limits on banking entities' investments in **hedge funds and private equity**.

1.4.1.4 Singapore Singapore aligned with the G20 recommendations through increased standardisation of derivative contracts, increased central clearing, more reporting to trade repositories, and higher capital requirements for non-centrally cleared contracts reflecting their greater counterparty risk.

In adopting a clearing regime, **MAS** considers three factors:

- **a.** The level of systemic risk posed by the product.
- **b.** The depth and liquidity of the market for the product.
- **c.** The availability of OTC CCPs regulated by MAS as **approved clearing houses (ACHs)** or **recognised clearing houses (RCHs)**.

From **1 October 2018**, MAS required all **SGD and USD fixed- and floating-rate swaps** to be cleared through MAS-regulated CCPs, to make derivatives trading safer and reduce counterparty risk. These are the more widely traded interest-rate derivatives in Singapore.

MAS began with the banks holding the largest share of OTC derivatives activity, to address the largest counterparty credit risks first. Banks whose **gross notional outstanding OTC derivatives exceed S**$20 $billion * *must clear through MAS - regulated CCPs; those banks accounted for * *90**20 billion** gross notional outstanding derivatives contracts booked in Singapore for each of the last **four calendar quarters**.

As in the US, EU and Japan, MAS **exempted intra-group transactions**: they do not transfer risk in or out of a corporate group. MAS also exempted **public bodies** — all central banks and governments, plus international multilateral organisations such as the **Bank for International Settlements, the International Monetary Fund and the World Bank** — because their policy functions may require flexibility that clearing obligations would constrain.

Clearing of interest-rate swaps matters because IRS constitute **more than 90% of interest-rate derivative contracts booked in Singapore** by gross notional outstanding, so extending clearing to them significantly reduces systemic risk in the Singapore financial system.

Clearing obligations are more enforceable where both counterparties book and clear in Singapore. They are harder to enforce for cross-border transactions where the foreign regulator does not require counterparties to seek central clearing; the interaction of jurisdictional rules remains an ongoing challenge. The **US and Japan** have commenced mandatory clearing; the **EU** has put a regime in place that has yet to commence.

For transactions traded in Singapore but booked elsewhere — foreign subsidiaries or local branches of foreign banks — MAS maintains oversight through **consolidated supervision** of local banking groups, engages banks regularly on their booking policies and risk controls as part of banking supervision, and assesses risk from **trade data reported to licensed trade repositories**, which includes OTC contracts booked or traded in Singapore.

With the global step-up in regulatory requirements and cost, banks became noticeably more selective about derivatives business, systematically scaling down investment banking activities including proprietary trading, capital-intensive activities and higher-risk exotic and credit derivatives **from 2008 to 2016**. In **2019** OTC derivatives trading recovered to a degree, growing more rapidly than exchange-traded derivatives.

1.4.2 Markets in Financial Instruments Directive II

MiFID II is a set of foundational regulations and directives governing capital markets in the **European Union**. EU-domiciled banks and their branches are subject to it, and so are banks that deal with EU investors, individual or corporate. Singapore hosts many branch operations of EU financial institutions that must meet MiFID II guidelines.

Singapore brokerage firms and local banks are not bound by the EU directive, but they are likely to be called on to support the compliance processes of their European counterparties, which must report a wide range of transactional data. As good corporate citizens in a global financial centre, Singapore institutions and exchanges can be expected to assist EU banks in meeting those requirements.

Most of the review burden falls on senior managers and compliance teams. Much of the additional workflow and documentation should be automated in systems, which requires both operations and sales staff to ensure verification procedures and data are conducted and recorded in line with MiFID II. An investment professional should know their organisation's obligations, or its decisions on alignment, and their own role under the new rules.

Role or activity
You execute a trade for a local client involving the purchase of EU securities (e.g. BMW, LVMH) from ar
You have EU clients — for example, a local fund manager working for a private bank with EU-based ultra-high-net-wo
You work for an institution that is a market maker for several EU-listed securities and have a trading des
You distribute EU funds in Singapore to APAC clients.
You work for a subsidiary of an EU institution conducting both underwriting and secondary placement.

A Singapore or APAC distributor must not assume that a foreign regime displaces local obligations: product governance, KYC, suitability, disclosure, conflicts and Singapore licensing all continue to apply.

1.5 Structured Products

Structured products combine one or more derivatives with an underlying investment. The underlying can be a single security, a group or basket of securities, or an index. They are popularly sold as **structured deposits, structured notes or structured funds**, offering investors the chance to optimise yield or participate in movement of the underlying.

Structured products are often OTC and are designed to create greater yield for securities the client already holds, or potential participation in the movement of an underlying security, index or basket. Different combinations and outcomes are designed to attract investors, each with its accompanying risks.

In exchange for returns better than the market, investors bear the risk of having to **convert to, or own, securities** if the underlying reaches — or as long as the price does not reach — a particular trigger level. The structure often requires the investor to have **sold put options** to the counterparty, which can result in accumulating more of the underlying as it falls. Investors must be comfortable bearing those inherent risks. Structures can also involve **leverage**, which magnifies both returns and losses.

1.6 The Singapore Derivatives Market

Singapore is a major hub for derivatives trading in Asia. **SGX-DT**, the derivatives arm of SGX, is a Pan-Asian derivatives exchange and the world's most liquid offshore market for the benchmark equity indices of **China, India, Japan and Taiwan**. It is the leading commodities exchange for contracts on **iron ore and rubber**, and provides **Eurodollar, Euroyen, Japanese government bond and stock index** futures trading for Asian investors. It also offers interest-rate futures for trading and hedging — including hedging loans and deposits and interest-rate swaps denominated in **SGD, USD and JPY** — plus intra-commodity, inter-commodity and inter-market trading and arbitrage services.

SGX exchange-traded derivatives volume and revenue, 2015–2019, show contracts rising from 161.2 million to 240.3 million with revenue between SGD 295.7 million and SGD 459.7 million; volume rebounded in the last two years of that period after falling in 2017.

Singapore is the leading FX hub for Asia and the **third largest in the world after New York and London**, benefiting from the synergy of a large FX derivatives business. It lags the major New York and London exchanges in other asset classes. The commencement of two further derivatives exchanges — **ICE Futures Singapore** and **Asia Pacific Exchange (APEX)** — added breadth to the Singapore market.

The **SFEMC** October 2020 semi-annual survey reported average daily turnover in OTC foreign exchange derivatives of **US$91 billion** in FX swaps and options, a decrease on the previous period that coincided with the global downturn caused by the COVID-19 pandemic. Use current statistics for present-day decisions.

1.7 Summary

- A derivative is a financial instrument whose value is linked to the value or volatility of an underlying reference or asset. Derivatives trade on centralised regulated exchanges or bilaterally in the less regulated OTC market.
- Derivatives and structured products let investors hedge, take a directional view without owning the asset, augment profits, change the payoff profile, profit from a fall, secure a right without obligation, arbitrage, change the nature of an asset or liability, and create synthetic positions.
- Futures, forwards, swaps, options and warrants are the basic or "plain vanilla" derivatives. Exotic derivatives combine types or add features. The building blocks are FX, money market interest-rate instruments, fixed income, equities, commodities, and mixes or indices of these.
- Participants use derivatives to hedge and manage risk on existing positions, to speculate, or to capitalise on perceived arbitrage between assets and markets.

Exam cues

- **Five building blocks:** FX, money market, fixed income, equities, commodities. Indices, baskets and funds are extensions, not a sixth class.
- **Numbers:** money market instruments are **>95%** of global exchange-traded futures and options and **>3/4** of OTC trades 2015–2019; maturities **1 day to 1 year**, typically **<90 days**. FX derivatives are the **second largest** category. OTC traded **>5×** exchange-traded value in 2018.
- **Regulatory dates:** EMIR **2012**; UK ring-fencing Act **2013**; Dodd-Frank **2010**, Volcker conformance **21 July 2015**; MAS mandatory clearing **1 October 2018** for SGD and USD fixed/floating IRS, threshold **S$20bn** gross notional over the last **four** quarters, covering banks holding **90%** of OTC contracts; IRS is **>90%** of SGD-booked interest-rate derivatives.
- **VIX:** the CBOE VIX measures perceived S&P 500 volatility from option prices. On **5 Feb 2018** the VIX more than doubled in a day; **SVXY** and **XIV** fell **>80%**, and XIV hit a termination event. ProShares later halved SVXY/UVXY leverage.
- **MiFID II** reaches Singapore desks: T+1 reporting, **15-minute** post-trade transparency, retail/professional/elective-professional classification, target market, inducement disclosure. A foreign regime does not replace Singapore obligations.
- **Trap:** ETFs and ETNs are **not** derivatives, even where they use futures and options internally. And shorting a put is **not** a hedge against downside risk.

Chapter 2 — Futures, Forwards & Swaps

Futures and forwards are the oldest derivatives, and swaps are built from repeated forwards. This chapter covers their mechanics, pricing, settlement and the four swap families the examination tests.

Learning Objectives

The candidate should be able to:

- Discuss the origins and evolution of futures and futures exchanges.
- Explain the economic importance of futures markets.
- Know the difference between futures and other assets and financial instruments.
- Describe the general features of futures contracts.
- Explain the differences between futures and forward contracts.
- Explain specific items within the contract specification of a futures contract.
- Explain and carry out basic calculations on notional value and margin requirements.
- Describe the different types of financial and commodity futures.
- Explain the role of futures exchanges and how they operate.
- Describe the settlement process of futures exchanges, including SGX-DT.
- Show how leverage can impact the performance returns of futures.
- Explain futures pricing models and how arbitrage opportunities arise.
- Explain and apply the concept of basis in futures contracts.
- Explain how to use interest rate swaps, FX swaps, currency swaps, equity swaps and credit default swaps.

2.1 Introduction — Futures and Forwards

Futures are standardised contracts for the purchase and sale of financial instruments or physical commodities on regulated futures exchanges, with the price determined for future delivery. Originally these were non-homogeneous contracts made directly between buyers and sellers — **forward contracts**. As particular specifications of quality, size and delivery time became widely used, the more popular terms standardised. With the desire for a credit-worthy market-maker or counterparty, margin requirements were set in and the futures exchange was born.

A futures exchange is a marketplace for standardised financial and commodity derivatives where buying and selling is contracted for future delivery.

Exchange	Position
CME Group	The world's largest and most diverse derivatives exchange: **3 billion contracts**
ICE (Intercontinental Exchange)	Second largest, built through mergers and acquisitions among exchanges and clea
ICE Futures Singapore (IFSG)	Created from ICE's acquisition of the **Singapore Mercantile Exchange (SMX**
APEX	Asia Pacific Exchange Pte Ltd, a Chinese-investor-backed exchange, granted MAS
Eurex	The biggest futures exchange in Europe, owned by Deutsche Börse.
Euronext	The second largest in Europe, offering access to Belgium, France, the Netherlands
B3 S.A.	Formerly BM&F Bovespa, of São Paulo — the largest futures market serving Sou
SGX	Operates Asia's largest derivatives market in futures, swaps and options, benefitin
HKEX	In addition to its equity and futures exchange, Hong Kong Exchanges and Clearin
Chinese exchanges	The Shanghai Futures Exchange, Dalian Commodities Exchange and Zhengzhou
Others	Japan Exchange Group, Korea Exchange, National Stock Exchange of India, Mos

Given the interconnectivity of today's markets, investors reach futures through multiple platforms and service providers — banks, broking firms and internet-based brokers.

2.1.1 Futures versus Forwards

Forward contracts preceded futures. The differences decide which instrument fits a given exposure.

Futures
Standardised as to quality, quantity, delivery time and place.
The exchange is counterparty, so there is **no counterparty risk between the actual buyer and seller** — it is realise
Marked to market at the end of each trading day; participants must cover margin shortfalls under exchange rules.
Usually offset (closed out) on the exchange before maturity; live contracts settle on the settlement price and procedures a
Globally, most trading now occurs on electronic platforms linked to the exchanges, with pit trading phased out.

Forwards remain more useful where a player needs a customised solution — a position or hedge in an exotic currency, or a delivery period not readily available on futures. Those needs are met through a customised contract with a financial institution or another commercial counterparty.

Understanding both matters because the directional and leverage effects of futures let an investor express the degree of bullishness or bearishness they hold.

2.1.2 Economic Importance and Impact of the Futures Market

The futures market is highly active and central to the global financial marketplace. It is an important source of market information and often indicates market sentiment.

2.1.2.1 Price Discovery The futures market reflects the current price of a contract for a product to be delivered in the future, providing a platform for ongoing **price discovery** — buyers and sellers contracting and influencing prices now, for future delivery, given available information and predictions. How participants absorb and react to information constantly changes prices. That information includes future economic performance, potential political developments, and forecasts of environmental and climatic change that may or may not come about.

2.1.2.2 Risk Management The futures market is where investors manage risk. Because market prices are transparent and readily available, investors know how much they need to buy or sell. For consumers and buyers of commodities, hedging lowers the eventual cost to consumers by reducing the risk that manufacturers raise prices for key supplies because of spot-market volatility. Business managers and financial investors use suitable futures as part of planning to reach their objectives.

2.1.2.3 Augmenting Volatility Participants use futures to leverage long or short positions. In major financial distress in equity, bond or commodity markets, the leveraging impact of futures and options can significantly augment volatility. The downward pricing pressure from shorting futures usually adds selling pressure on the spot market and underlying assets; hedge funds and traders exploit such situations, and the value of underlying assets can fall faster than normal, destroying market value over a very short time.

Computer-driven trading and electronic platforms have quickened the pace of wealth destruction during shocks and extended its ripple into other asset classes and the economy, so governments have had to intervene in both futures and spot markets in extreme situations to dampen that ripple.

2.2 General Features of Futures Contracts

- **a. Standardised.** Futures are standardised contracts for the purchase and sale of financial instruments or physical commodities on a regulated futures exchange, for future delivery on the delivery date.
- **b. Price driven by the underlying.** Price movement is directly influenced by the underlying asset's price and value. The futures price is set by current demand and supply, which are in turn shaped by composite information and market expectations about the future.
- **c. Finite life.** A contract ends on the expiration date, so market direction and time horizon are central considerations.
- **d. Notional value.** Futures carry a notional value depending on contract size or multiple and the market price at any time. At expiration the parties must fulfil their obligations — one to deliver, the other to receive. Unlike out-of-the-money options or warrants, which go to zero at expiry, a futures contract **retains a notional value** at expiration.
- **e. Lower counterparty risk.** The counterparty is the futures exchange, backed by margin requirements.
- **f. Leverage.** The investor posts only a fraction of contract value as initial margin to gain exposure to the whole position, augmenting both gains and losses.
- **g. Daily mark-to-market.** The value is marked at the end of each trading session and any margin shortfall must be covered.
- **h. Close-out usual.** In most executions buyers and sellers do not deliver or receive the underlying, but square off with the exchange to realise gain or loss.
- **i. Contract-specific terms.** Contract size and units depend on the type of futures; they appear in the contract specification or term sheet. Traders combine futures characteristics with the underlying and other derivatives to craft strategies with different payoff and risk profiles.

2.2.1 Calculating Notional Value for Different Types of Futures Contracts

Notional value depends on the futures price and the multiplier for that contract:

$$\text{Notional value} = \text{futures price} \times \text{multiplier (contract size)}$$

The multiplier differs by contract, and by exchange for the same underlying index.

Contract	Setting	Calc
E-mini S&P 500, Jun 2020	Index at 2,358.50 on 31 March 2020; bid/ask 2,361.00 / 2,361.25; multiplier 50	Purc
E-mini Nasdaq 100, Jun 2020	Contract done at 5,429.25 on 4 April; multiple 20	20 ×
SGX FTSE China A50, Sep 2020	Trading at 10,300 on 4 April; multiple 1	1 × 1
EUR/USD futures, Jun 2020	Last traded at 1.07045; contract unit EUR 125,000	EUR

2.2.2 Leverage

- **a. Initial margin.** The small upfront payment that initiates a position; the investor does not pay for the whole contract at trade initiation.
- **b. Maintenance margin.** The minimum the customer must keep on deposit at all times. While net equity is above it, no request for additional funds arises even after an adverse move.
- **c. Additional margin.** Each account is credited or debited at the end of each trading day and checked against the appropriate margin for all open positions. If the balance falls below maintenance level, additional margin is required and the account must be restored to the **initial** margin level immediately or within a stipulated time.

In Singapore, **SGX** publishes maintenance and initial margin for every futures contract on a daily margin schedule. Margins vary when market conditions change significantly — for example, in extreme volatility.

How margins are used: SGX Nikkei 225 The contract is worth **JPY 500 × the Nikkei level**. At 22,500 the contract value is:

$$500 \times 22{,}500 = \text{JPY } 11{,}250{,}000$$

At an initial margin of **6.5%**:

$$0.065 \times 11{,}250{,}000 = \text{JPY } 731{,}250$$

If the index rallies 7% to 24,075, the contract is worth JPY 12,037,500. The profit is:

$$500 \times (24{,}075 - 22{,}500) = \text{JPY } 787{,}500$$

That is more than **100% of the initial margin** — the leverage effect, and one reason futures appeal.

If the index falls 160 points to 22,340, the contract is worth JPY 11,170,000 and the loss is:

$$500 \times (22{,}340 - 22{,}500) = -\text{JPY } 80{,}000$$

The exchange compares contract value to the account daily and adds profits or subtracts losses. The balance must stay above the maintenance margin — **5%**, or:

$$0.05 \times 11{,}250{,}000 = \text{JPY } 562{,}500$$

Here the balance becomes JPY 651,250, still above the JPY 562,500 maintenance level.

2.2.3 Contract Specifications and Term Sheets

A contract specification sets the basic terms and conditions of the investment. It should state:

- description and standard size or denomination of the underlying;
- contract size and value;
- tick size;
- trading hours;
- price limits;
- last trading day;
- settlement basis;
- final settlement price;
- mark-to-market procedures;
- margin call procedures.

Appendix C lists the main SGX-DT contract specifications.

Term	Meaning
Contract standardisation	Each contract has a standardised size fixed in currency and contract size relative to the un
Contract value	Contract size × current price. At an index level of 1,850, one E-mini contract is USD 50 ×
Tick size	The smallest amount the contract price can move; it varies by contract. An E-mini tick is
Price limits	The maximum the price may move in one day from the previous settlement. Set by the ex
Mark-to-market	Contracts are marked daily as a safety measure. At the end of each day the exchange sets
Margin call	Accounts are credited/debited on the day's profit/loss and checked against open positions.

The SiMSCI futures contract

Term	Specification
Contract size	SGD 200 × Singapore MSCI (SiMSCI) futures price
Contract months	2 nearest serial months and 4 quarterly contract months
Trading hours	8.30 am – 5.15 pm (T session); 6.15 pm – 10.55 pm (T+1 session)
Minimum price fluctuation	0.1 index point, worth **SGD 20**
Daily price limits	On a **15%** move either way from the previous settlement, trading at or within ±15% is allow
Last trading day	The **second-last business day** of the contract month. A business day is one on which the
Settlement basis	Cash settlement
Final settlement price	The SiMSCI value computed on the **Special Quotation** methodology applied to each comp

2.3 Types of Futures

Growth in futures began with farmers, miners, manufacturers and businesses needing to manage risk and uncertainty: a futures contract fixes a price for a period, which makes planning easier. The categories are:

- **a. Commodity futures** — agriculture, energy, metals.
- **b. Financial futures** — equity, interest rates, foreign exchange.
- **c. Others** — innovation has created futures in new areas such as weather.

2.3.1 Interest Rates Futures

Banks lending or borrowing over a period are exposed to rate movements and can partially or fully hedge through interest-rate futures or options on them. More broadly, these contracts let investors and traders take positions on how short-term or long-term rates may move across the yield curve, given fiscal and monetary policy and their view of economic statistics. Interest-rate futures are available on most exchanges, especially in the US, UK and Singapore.

2.3.1.1 Short-Term Interest Rates Futures The most popular short-term contracts are **US Treasury bill futures, Eurodollar futures and Euroyen futures**. The underlying is typically:

- **a.** A time deposit in a **Eurocurrency** — a currency lent or borrowed outside its country of origin — where the rate offered for a specified period to a first-class bank determines the futures price for the same period; or

- **b.** A government treasury bill, where the rate at which an investor is willing to invest for the period forms the basis of the futures price.

$$P = 100 \times (1 - R)$$

where

$$R$$

is the annualised interest rate for the period. At

$$R = 3.50\%$$

:

$$P = 100 \times (1 - 0.035) = 96.50$$

2.3.1.2 Long-Term Interest Rates Futures Used to manage rate exposure or to take a market view. A borrower with outstanding debt may protect against rising rates; a fixed-income manager may sell interest-rate futures to protect a bond portfolio. If rates rise, futures prices fall — rates and bond prices move in opposite directions — and the profit on closing the futures offsets the portfolio decline.

$$\left(1 + \frac{d_1 r_1}{360}\right)\left(1 + \frac{d_2 r_2}{360}\right) = \left(1 + \frac{DR}{360}\right)$$

Rearranged to solve for the second-period rate:

$$r_2 = \frac{360}{d_2}\left(\frac{360 + DR}{360 + d_1 r_1} - 1\right)$$

where

$$d_1$$

is days in the first period,

$$d_2$$

days in the second,

$$D$$

days for the entire period,

$$r_1$$

the first-period rate,

$$r_2$$

the second-period rate and

$$R$$

the rate for the whole period.

Worked example. Spot date to 15 September is 45 days; to 15 December is 135 days. The 45-day rate is 1.25% and the 135-day rate is 1.75%. The implied 90-day forward rate is:

$$\left(1 + \frac{45 \times 0.0125}{360}\right)\left(1 + \frac{90 \times r_2}{360}\right) = \left(1 + \frac{135 \times 0.0175}{360}\right) = 1.006563$$

Solving gives

$$r_2 = 2.00\%$$

.

2.3.1.3 US Treasury Futures US Treasuries are among the most liquid financial products in the world, and Treasury futures and options support hedging, income enhancement, duration adjustment, rate speculation and spread trades. CME contracts mirror the **2, 5, 10 and 30-year** tenures. On SGX, **mini-Japanese Government Bond futures** are the popular long-bond contract.

Each US Treasury futures contract has a face value at maturity of **USD 100,000**, except the **2-year and 3-year** contracts at **USD 200,000**. Prices are quoted in points per **USD 2,000** for 2- and 3-year contracts and per **USD 1,000** for all others.

Profit example. A fund manager expects Treasury yields to rise. He sells 10 contracts of Jun 2017 5-year Treasury Note futures at 117.245 and buys them back at 116.000:

$$10 \times (117.245 - 116.000) \times 1{,}000 = \text{USD } 12{,}450 \text{ profit}$$

The 5-year tick is **¼ of 1/32** of a point, worth:

$$0.25 \times 0.03125 \times 0.01 \times 100{,}000 = \text{USD } 7.8125$$

so the gain is

$$(117.245 - 116.000) \times 1{,}000/7.8125 = 159.36$$

ticks, and

$$10 \times 159.36 \times 7.8125 = \text{USD } 12{,}450$$

.

Loss example. On Thursday 7 April 2017 a surprise US missile strike on Syrian air bases pushed yields down and bond prices up; the 10-year yield fell to 2.31%, the lowest since the March rate hike when it was 2.6%. A trader reads the June 2017 10-year T-Note futures as having risen only modestly, from 125 8/32 to 125 12/32, and buys 10 contracts at 125 12/32 expecting more buyers. Prices reverse and the contract ends the day at 124 8/32:

$$10 \times (124\tfrac{8}{32} - 125\tfrac{12}{32}) \times 1{,}000 = -\text{USD } 11{,}250 \text{ loss}$$

The 10-year tick is **½ of 1/32**, worth USD 15.625, so the loss is **−72 ticks**, and

$$10 \times -72 \times 15.625 = -\text{USD } 11{,}250$$

.

2.3.2 Currency Futures

Currency futures are contracts to buy or sell a specified amount of one currency against another at a set price and future date. Financial institutions, investment managers, corporations and private investors use them to manage currency risk or to profit from rate movements. They were first introduced at the **Chicago Mercantile Exchange in 1972**, after the fixed exchange-rate system and gold standard were discarded.

Currency futures volume tracks the underlying spot market. The BIS Triennial Survey reported global FX turnover of **USD 5.1 trillion per day** in 2016, **5.5% below** USD 5.4 trillion in 2013 (the source text prints "5.1 billon"; read trillion). Part of the decline reflected lower growth in global trade and capital flows, still below pre-crisis levels.

More significant was a structural shift towards using FX for hedging rather than risk-taking, a consequence of reduced proprietary trading and tighter risk controls. Banks consolidated and cut costs over the decade. The number of dealer banks willing to warehouse risk declined while **non-bank market-makers** gained a stronger foothold as liquidity providers, sometimes bypassing traditional bank intermediaries. Despite the overall decline, trading in most FX derivatives — particularly FX swaps — kept growing, and emerging-market currencies, especially the **renminbi**, gained share. Singapore kept the top spot for FX trading, which supports FX-derivative business here.

2.3.2.1 Types of Foreign Exchange Futures Contracts Offered Futures activity mirrors demand in the underlying currency market, so major pairs dominate. On the **CME**, currency futures are grouped into majors against the USD, emerging-market currencies against the USD, cross-rates not involving the USD, and **E-Micro** contracts trading at **1/10th** the size of the regular contract. On **Euronext**, trading concentrates on the more active majors — **EUR/USD, GBP/USD and GBP/EUR**.

On **SGX-DT**, Asian currencies trade against the USD, among themselves, and against the non-deliverable forward for the **renminbi (CNY), Indonesian rupiah (IDR), Taiwan dollar (TWD), Indian rupee (INR), Korean won (KRW), Malaysian ringgit (MYR) and Philippine peso (PHP)**. SGX remains one of few venues offering hedging, trading and investment exposure in these currencies, and Singapore's FX derivatives market benefits from demand for emerging Asian currencies as China, Indonesia and India expand.

2.3.2.2 Interest Rate Parity Theory The theory holds that the forward (futures) premium or discount between two currencies should equal the difference in domestic interest rates for securities of the same maturity, ignoring small transaction costs. Where the relationship is violated, arbitrageurs can earn risk-free profit and their actions force futures and spot rates back into line.

$$F = S \times \frac{1 + R_c\left(\frac{n}{360}\right)}{1 + R_b\left(\frac{n}{360}\right)}$$

where

$$F$$

is the forward (futures) rate,

$$S$$

the spot rate,

$$R_b$$

the annualised base-currency rate,

$$R_c$$

the annualised counter-currency rate and

$$n$$

the number of days.

Worked example. Base currency GBP, counter currency USD. Spot GBP/USD = 1.2422;

$$R_b$$

= 1-year GBP LIBOR = 0.70444%;

$$R_c$$

= 1-year USD LIBOR = 1.79900%.

$$F = 1.2422 \times \frac{1 + 0.01799\left(\frac{360}{360}\right)}{1 + 0.00704\left(\frac{360}{360}\right)} = 1.2557$$

So GBP 1 = USD 1.2557, a forward price of GBP/USD 1.2557.

2.3.3 Equity Index Futures

Equity index futures replicate the future performance of an underlying equity market index, letting investors capitalise on broad market movements without changing their dollar investment in stocks. An index is a representative basket or portfolio of stocks or commodities; an equity index measures the performance of a selected basket representing the whole market. Indices may be **price-weighted** (DJIA) or **market-capitalisation weighted** (MSCI).

Buying the index raises or lowers exposure to specific markets or sectors and is a cost-effective way to gain equity exposure. Most exchanges list futures on their own indices, while some position themselves regionally

or globally. SGX has consolidated its position as a regional platform for futures, options and structured warrants on Asian markets, and offers index futures relevant to managers benchmarked to indices such as MSCI, so they can rebalance rapidly, hedge positions and exploit country-specific opportunities. The most successful launch has been the **SGX FTSE China A50 Futures**, which continues to trade heavily.

SGX equity index futures include **MSCI Singapore, Straits Times Index, FTSE China A50, FTSE China H50, FTSE Malaysia, FTSE Philippines, FTSE Taiwan, FTSE Thailand, Nikkei 225, USD Nikkei 225, Mini Nikkei 225 and Nifty 50**.

2.3.3.1 Differences between Equity Index Futures and Stocks

Feature	Equity index futures	Stocks
What is owned	A cash-settled contract on the index	Legal ownership of shares
Settlement	Daily mark-to-market; cash settlement at expiry	T+2 settlement; no daily margin
Leverage	Margin only	Full consideration, or margin financing
Short selling	Direct, without borrowing stock or an uptick rule	Requires stock borrow, with recall and cost risk
Income	No dividend entitlement	Dividends and voting rights

2.3.3.2 Pricing Equity Index Futures

$$F = S\left[1 + r\left(\frac{t}{360}\right)\right] - D$$

where

$$S$$

is the spot index,

$$r$$

the short-term rate,

$$t$$

days to expiry and

$$D$$

dividends paid before expiry.

Index futures do not trade precisely at fair value all the time; the difference between futures and spot is the **basis**. Fair value is normally expected to be positive — futures above spot — because under normal conditions financing costs, reflected in rates such as LIBOR or SIBOR, exceed dividend yields. Loose post-2008 monetary policy has at times reversed this.

If futures rally much above fair value, an arbitrageur buys the equity portfolio and sells index futures, trading a basket or subset of constituents or using electronic systems to trade all or nearly all of them. Buying stocks and selling futures bids stocks up or pushes futures down until equilibrium returns.

2.3.4 Single Stock Futures

A **Single Stock Futures (SSF)** contract tracks the price of the underlying stock. On SGX the buyer agrees to pay a specified price at a predetermined date and the seller is obligated to deliver the value of the stock at that date; SGX SSFs are **cash settled**, with no physical delivery. Portfolio managers use them to protect against adverse moves and to diversify risk using margin financing. SSFs carry **no shareholder privileges**.

Advantages:

- **a.** Leveraged, based on several thousand shares, so the leverage effect amplifies gains and losses.
- **b.** Lower transaction cost than buying the stock, because the outlay is only the margin.
- **c.** For the same reason, offshore investors bear less currency exposure — they invest only the margin, not the full underlying value.
- **d.** Ease of short selling lets investors participate in and profit from market downturns.
- **e.** Once an exchange admits a contract it must ensure liquidity, transparency, fairness and timeliness in market-making with reasonable bid-offer spreads, and investors must be able to close out, typically electronically. The exchange guarantees clearing-house support from designated market-makers.

The main disadvantage is that SSFs can invite short selling of the stock, reducing underlying prices and causing downward spirals in uncertainty and shocks. The market's ability to absorb increased downward volatility is an important consideration before a new contract is introduced.

2.3.5 Commodity Futures

Physical-market storage, transport, quality and delivery-location differences create basis and roll risk. Commodity contracts are grouped as agriculture, energy and metals.

2.3.5.1 Gold and Other Precious Metal Futures In potential military conflicts involving major powers there is a tendency to **flight to safety**. In early April 2017, tension escalated between the US and Syria–Russia over banned chemical weapons and the US missile response, while US–North Korean nuclear-testing tension also rose. Gold futures moved up **4% to USD 1,290.70/ounce** just before Easter; when tension eased the price retraced.

Calculating notional profit for gold futures On 10 April 2017, with US military tension involving Syria, North Korea and Russia, June gold gained 0.39% on escalating concern. At the start of European trading, Jun 2017 gold futures on ICE traded at **USD 1,259.10/Troy ounce**. A trader expecting escalation bought **100 contracts**, each **100 Troy ounces**. Gold rose to USD 1,288.50/ounce before Good Friday, 13 April.

$$100 \times (1{,}288.50 - 1{,}259.10) \times 100 = \text{USD } 294{,}000$$

Conversely, when interest rates rise, gold and other precious metals come under pressure because they yield no interest or dividend; when perceived risk is low, investors prefer risky assets and safe havens fall. Gold is also increasingly intertwined with **gold ETFs**: according to the World Gold Council the largest growth in gold demand in **2016 (+660%)** came from ETFs.

Beyond financial markets, gold has jewellery demand: **China and India** are the largest sources, and more than half of physical gold trading is estimated to come from Asia. Interruption of precious-metal imports creates underlying downward pressure.

Gold futures on the **CME and ICE** are quoted in **100 troy ounces**. In Singapore the **SGX Gold contract is dormant** and not traded.

2.3.5.2 Base Metal and Ferrous Metal Futures Demand and prices depend on economic growth, with underlying demand from major traders, miners and manufacturers wanting physical delivery. Base and ferrous metals rose significantly during China's infrastructure expansion, especially **2002–2012**; China's slowdown **2012–2016** amid overcapacity produced a sharp fall. Expected demand from proposed US infrastructure fiscal spending in **2017–2018** lifted prices again.

Steel pricing is not uniform and US–China trade disputes and anti-dumping allegations disrupt price discovery. The US announced **steel and aluminium tariffs** to protect local manufacturers, which widens the difference between US and non-US exchange prices. Steel comes in different types and grades, and SGX steel quotes from China differ from CME quotes featuring US steel.

Iron ore and steel are key SGX contracts. The **London Metal Exchange** is the largest global metal exchange and remains a main venue for iron ore and steel futures, while Chinese exchanges have grown strongly. Singapore is positioned as a key **clearing centre for iron ore** traded elsewhere, especially China.

SGX iron ore and steel contracts:

- **a.** SGX TSI Iron Ore CFR China (62% FE Fines) Index
- **b.** SGX HRC CFR ASEAN Steel
- **c.** SGX Platts Iron Ore CFR China (Lump Premium) Index
- **d.** SGX MB Iron Ore CFR China (58% FE Fines) Index
- **e.** SGX TSI Iron Ore CFR China (58% FE Fines) Index

2.3.5.3 Crude Oil Futures Crude oil futures remain a focal point for investors because of dramatic price volatility: a high of **USD 155/barrel** just before the Global Financial Crisis, a plunge to **USD 48** in 2009, a rebound above **USD 100** for most of 2010–2013, then new lows of **USD 29.60** in 2015 amid structural change and competition from fracking. Futures may have helped producers lock in higher prices on some

occasions, but extended declines offer producers little protection. An **OPEC** agreement to limit supply eventually held crude in a **USD 55–65** range in the later part of 2018, ahead of the Saudi Aramco listing. Crude remains one of the most actively traded instruments.

2.3.5.4 Energy — Natural Gas, Other Refined Products and Coal Futures The volatility of energy futures is relatively well correlated. Like crude, other traditional energy products — **LNG, jet fuel, gas oil, kerosene, coal** — have been volatile over the decade. For downstream refined products there may be a **lag effect** before the full impact of price changes works through inventory. Coal, though environmentally unattractive, remains a viable substitute in many developing economies.

Singapore is a major petrochemical refining base, and SGX lists futures on diverse petrochemical and refined products from regional players, plus **Indonesian coal** futures.

2.3.5.5 Energy — Electricity Futures Energy futures have been available in the US for many years. Multiple production sources (LNG, coal, hydroelectricity) across regions, many users (towns, cities, industries), and the need to optimise non-peak hours against fluctuating demand drove the development of electricity futures.

On **SGX** there are monthly and quarterly electricity futures, still relatively new and serving specific needs of Singapore industrial users. The local market is thermal, with **more than 90% dependent on natural gas**. Generating companies offer to sell into the spot market **every half-hour**, matching prevailing demand, but prices fluctuate during tight supply from plant outages or network constraints. Consumers buying directly from the spot market are exposed to those fluctuations. Electricity futures offset expensive electricity for consumers or secure higher proceeds against cheaper off-peak electricity.

2.3.5.6 Agriculture Futures A wide range of agriculture contracts — **grain, soft commodities, livestock, dairy and forest products** — has existed for decades, driven mainly by the supply and business needs of farmers, producers, wholesale consumers, investors and speculators.

2.4 Contract Settlement and Clearing Process and Orders on the SGX

2.4.1 Modes of Settlement

A buyer or seller can settle a futures position in three ways.

2.4.1.1 Delivery or Hold to Expiry Every contract has an expiry date. On that date the buyer accepts delivery of the underlying and the seller liquidates by delivering at the agreed price. Commodity contracts state physical delivery to an approved warehouse; currencies and some financial instruments such as the S&P 500 call for **cash settlement**. Every contract specifies the last trading day before expiry.

Investors must know the expiry dates, because as expiry approaches **liquidity decreases** as traders roll to the next contract month.

2.4.1.2 Offset Position The simplest and most common route for financial futures. Buying two S&P 500 contracts is offset by selling two; selling two is offset by buying two. To limit overnight risk, many traders exit all positions and are **flat** at the end of each trading day.

2.4.1.3 Roll Position Rather than closing out, the investor closes the expiring contract and simultaneously opens a position in a later month, maintaining exposure and deferring settlement.

2.4.2 Minimum Price Movement for Contracts on SGX-DT — Understanding Tick Sizes

The tick size is three things at once: the minimum movement in the quoted price, the minimum increment between bid and offer, and a dollar amount per contract. Convert between them using the contract multiplier.

2.4.3 Listing of Derivatives Contracts on SGX-DT

SGX-DT lists five categories:

Category	Contracts
FX	USD/SGD, AUD/USD
Equity indices	SiMSCI, Nikkei 225 and the wider Asian roster
Interest rates	Eurodollar, Euroyen LIBOR
Commodities	PLATTS Singapore Fuel Oil 380cst, SICOM TSR20 Rubber
Dividend indices	Nikkei Stock Average Dividend Point Index

2.4.4 Derivatives Clearing Process on SGX-DT

SGX-DC clears:

- **a.** SGX-DT products;
- **b.** OTC commodity trades; and
- **c.** OTC financial derivatives, through trade registration.

Margin cycles run **three times intraday** — late morning, afternoon and post-end-of-day. Settlement covers variation settlement, option premiums, OTC coupons, price alignment interest and clearing fees. **ICE Clear Singapore** clears IFSG products.

2.4.5 Mutual Offset System

Since **1984**, the CME mutual-offset arrangement lets positions opened on one exchange be closed on the other, facilitating global clearing. It applies to specific SGX products — **NK, NU, FEM and FCH** — and is product-specific, not general.

2.4.6 Potential Role of Singapore as a Global Clearing Centre

SGX-DC is a **CFTC-designated clearing organisation**, has been assessed for EU equivalence, and **ESMA** recognises SGX-DC and CDP. Combined with the CME mutual-offset links, this positions Singapore as a high-quality global clearing centre for futures and derivatives; as technology changes how transactions are performed, the clearing process must continue to adapt.

2.5 Futures Pricing Model

2.5.1 Spot Price

The spot price is the current market price of the underlying asset or commodity.

2.5.2 Cost of Carry Model

The relationship between futures and spot prices is summarised by **cost of carry**. Carry includes financial costs — interest on bonds, interest on margin accounts, interest on money borrowed to buy a security — and economic costs such as the opportunity cost of taking the initial position. The **net** cost of carry is all costs of holding the equivalent cash position less the returns that position would have generated: financing at prevailing rates, and dividends or bonuses from shares, or the average return of an index in the case of index futures. In futures the investor deposits only a fraction of position value as margin.

The model assumes arbitrage between cash and futures eliminates all pricing imperfections. For simplicity it assumes:

- **a.** the contract is held to maturity, so a fair price can be derived;
- **b.** there are no transaction costs or margin requirements;
- **c.** there are no restrictions on short selling; and
- **d.** investors can borrow and lend at the same rate.

Once arbitrage has removed distortions, the fair futures price is spot plus the net cost of carry from today to expiry:

$$F_{0,t} = S_0(1 + C_{0,t})$$

2.5.3 Cash-and-Carry Arbitrage

Position	Cash-and-carry (futures too high)	Reverse cash-and-carry (futures too low)
Debt	Borrow to buy the spot asset	Lend the proceeds of a short sale
Physical	Buy and hold the spot asset	Short or borrow the spot asset and deliver
Futures	Sell futures	Buy futures

Forward case. Gold spot is USD 1,275/oz, carry is 2.25% for the year, and the futures price is USD 1,305.

$$F_{0,t} = 1{,}275 \times (1 + 0.0225) = 1{,}303.69$$

Futures at USD 1,305 exceed fair value of USD 1,303.69, so the arbitrage is to sell futures and buy spot. Borrow to buy 100 oz:

$$1 \times 1{,}275 \times 100 = \text{USD } 127{,}500$$

Sell one 100 oz contract at USD 1,305 for USD 130,500. Repay principal plus interest:

$$127{,}500 + (0.0225 \times 127{,}500) = \text{USD } 130{,}368.75$$

Profit:

$$130{,}500.00 - 130{,}368.75 = \text{USD } 131.25$$

— a negligible **0.1%** opportunity. To improve it, the futures price must be higher, the spot lower, and the borrowing cost materially lower.

Reverse case. Spot gold is USD 1,285.

$$F_{0,t} = 1{,}285 \times (1 + 0.0225) = 1{,}313.91$$

Futures at USD 1,305 are below fair value of USD 1,313.91, so buy futures and short spot. Borrow 100 oz of gold at **1.00%** for a year and sell it at spot for:

$$100 \times 1{,}285 = \text{USD } 128{,}500$$

Lend the proceeds at **2.25%**, earning USD 2,891.25, so after a year:

$$128{,}500 + 2{,}891.25 = \text{USD } 131{,}391.25$$

Buy 100 oz one-year futures at USD 1,305 for USD 130,500 to cover the borrowed gold. Surplus:

$$131{,}391.25 - 130{,}500.00 = \text{USD } 891.25$$

Cost of borrowing the gold:

$$128{,}500 \times 0.01 = \text{USD } 1{,}285.00$$

Overall:

$$891.25 - 1{,}285 = -\text{USD } 393.75 \quad (-\mathbf{0.306}\%)$$

So there was **no arbitrage**. To create one, the gold borrowing cost must be much lower or free while the lending rate rises, or the futures price must fall or spot rise.

Arbitrage rarely exists or lasts more than milliseconds: computer-based trading has pushed efficiency past human levels and any differential disappears.

2.5.4 Expectancy Model

Futures prices are expected future spot prices. If more traders expect the futures price to rise than to fall, the current futures price is bid up. What matters is not the cash-futures relationship but the relationship between the **expected spot price at expiry** and the futures price. Both the cost-of-carry and expectations models explain part of futures price movement.

2.6 Basis in Futures Contracts

The difference between futures and spot is the **basis**.

When markets are in **contango** — typically stable or bullish conditions — the basis is usually positive, with futures above spot because of carrying costs such as financing and storage. In **backwardation** — often bearish or stressed conditions — the basis can be negative, with futures below spot.

Because a contract settles at the cash price on expiry, **futures and spot converge as expiry approaches**.

> **Convention warning.** The official guide states on page 46 that contango gives a *positive* basis with futures above spot, and then on page 47 defines and uses basis as *spot minus futures*, which makes contango's basis negative. The two are the same economic relationship described with opposite signs. This guide uses **basis = spot − futures** throughout, so contango produces a negative basis. State your convention in any answer.

2.6.1 Factors Affecting Basis

- **a.** Cost or return of carry.
- **b.** Time to maturity.
- **c.** Yield-curve changes.
- **d.** Relative liquidity of cash and futures markets.
- **e.** Market rates versus administered rates.
- **f.** Expectations of market participants.
- **g.** Coupon or income differences between the hedged asset and the deliverable.

2.6.2 Basis, Contango and Backwardation

Basis risk persists for three reasons: the cash and futures markets are not perfectly correlated, the exposure being hedged may not match the deliverable, and the hedge may be closed before expiry, so convergence is incomplete. A short hedger's effective price is the futures price plus the ending basis:

$$V_t = S_t + (F - F_t) = F + (S_t - F_t)$$

Roll yield arises when a position is rolled from an expiring contract to a later one; under contango the roll costs money, under backwardation it earns.

2.7 Swaps

Swaps are customised OTC agreements to exchange cash flows over several periods on a notional amount. Singapore's FX market is heavily swap-based: forex forwards and swaps were **70–75% of FX turnover in 2016–17**, and currency interest-rate swaps **60–66% of FX derivatives**.

2.7.1 Interest Rate Swaps

One party pays fixed and receives floating, or the reverse. The notional is normally **not exchanged**; the swap rate or spread and the benchmark reset dates matter.

Plain-vanilla example: USD 100 million, paying 5% fixed against six-month LIBOR. With LIBOR at 4.2% in September 2014, the fixed payer's net receipt is:

$$0.5 \times (0.042 - 0.05) \times 100\text{m} = -\text{USD } 0.4\text{m}$$

so Bank B receives USD 2.1m on the fixed leg and pays USD 2.5m floating. With LIBOR at 4.8% in March 2015 the floating leg is USD 2.4m and the net position reverses.

$$\text{fixed payment} = \text{notional} \times \text{swap rate} \times \frac{\text{days}}{360}$$

The **swap spread** is the swap rate less the yield on the government bond of matching maturity. The **LIBOR swap curve** plots swap rates by tenor. A party with an existing swap can enter an **offsetting swap** to reduce or cancel the exposure.

2.7.2 FX Swaps

An FX swap exchanges currencies on a near leg and reverses the exchange on a far leg, managing short-term currency funding. The near-leg rate is the spot rate and the far leg is set by the swap points for the value dates. Example: a firm needing EUR 5 million for six months sells USD for EUR spot and agrees to reverse the exchange in six months at the forward rate, locking the funding cost.

2.7.3 Currency Swaps

A currency swap exchanges **principal and interest** in two currencies. A Singapore bank needing USD swaps SGD principal with a US bank needing SGD, each paying the other's interest, transforming both the currency and the interest exposure for the life of the deal.

2.7.4 Commodity Swaps

A fixed price is exchanged for a floating commodity price, letting a producer or user lock or transform price exposure without repeatedly rolling futures. SGX lists standardised commodity swaps.

2.7.5 Equity Swaps

One party receives the **total return** of an equity or index and pays a fixed or floating rate, creating synthetic equity exposure without ownership.

Four-period example on a notional of USD 200 million against the S&P 500, where the manager pays the index total return plus 2% and receives 4% fixed:

Period	Index level	Index return	Swap cash flow (USD m)	Portfolio value (USD m)	Reported return
31 Jul 2017	2,470.20	—	—	400.00	—
Period 1	2,569.00	+4%	−2.00	414.00	+3.50%
Period 2	2,363.48	−8%	+10.00	390.88	−5.58%
Period 3	2,623.47	+11%	−9.00	424.88	+8.70%
Period 4	2,754.64	+5%	−3.00	443.12	+4.28%

The manager's realised return each period is the portfolio return plus the swap cash flow, which is why a falling index produces a positive swap flow.

2.7.6 Credit Default Swaps

A protection buyer pays a spread, usually in basis points quarterly, to a protection seller. On a defined **credit event** the seller pays under cash or physical settlement. Typical notionals are **USD 10–20 million**.

Premium arithmetic: at **160bp** on **USD 20 million**, quarterly:

$$160\text{bp} = 1.60\% \;\Rightarrow\; 20\text{m} \times 1.60\% = \text{USD } 320{,}000 \text{ p.a.}$$

so each quarter is one quarter of the annual amount, **USD 80,000** — equivalently 40bp of notional per quarter.

2.7.6.1 CDS Spreads The CDS "spread" is not the same as a benchmark spread; it is the premium for credit protection on the reference entity. Spreads widen with credit deterioration — the PIIGS (Portugal, Ireland, Italy, Greece, Spain) episode is the standard example.

2.7.6.2 Credit Event The ISDA-defined events are bankruptcy, failure to pay, restructuring, obligation acceleration, obligation default, repudiation/moratorium and (in some definitions) government intervention. Debt restructuring is the most negotiated of these.

2.7.6.3 Cash Settlement The protection seller pays the difference between par and the market price of the deliverable obligation after the event.

2.7.6.4 Physical Settlement The protection seller pays par and receives the deliverable obligation, so the buyer delivers the defaulted bond and receives face value.

Naked CDS creates large synthetic credit exposure. The AIG episode during the Global Financial Crisis showed how leverage, opacity and counterparty concentration amplify credit risk: AIG had sold protection far in excess of its capital and required a public rescue.

2.8 Summary

Futures are standardised, margined, daily-settled and cleared; forwards and swaps are usually customised OTC. Pricing is driven by carry and parity, but basis, liquidity, margin and credit constraints matter. Swaps exchange exposures rather than principal ownership and are powerful but documentation- and counterparty-dependent.

Exam cues

- **Notional** = price × multiplier. **Margin** = initial margin % × notional. Learn the Nikkei example: 22,500 × JPY 500 = JPY 11,250,000; 6.5% = JPY 731,250; a 7% index rise is a 107% return on margin.
- **Short-rate quote**
$$P = 100(1 - R)$$
, so rates up → price down. Long-rate pricing chains the two periods to the whole period and solves for
$$r_2$$
(the 45/135/90-day example gives **2.00%**).
- **US Treasury:** face **USD 100,000** except 2- and 3-year at **USD 200,000**; 5-year tick **USD 7.8125** (¼ of 1/32), 10-year tick **USD 15.625** (½ of 1/32).
- **Interest-rate parity:**
$$F = S \times (1 + R_c n/360)/(1 + R_b n/360)$$
; the GBP/USD example gives **1.2557**.
- **Arbitrage:** cash-and-carry when futures is **too high**, reverse when **too low**. The reverse gold case loses **USD 393.75 (−0.306%)** once gold borrowing at 1% is counted.
- **Basis = spot − futures.** Contango puts futures above spot, so under this definition contango gives a **negative** basis. Some texts define basis the other way round; state your convention before using it. Convergence forces basis to zero at expiry.
- **CDS:** 160bp on USD 20m is USD 80,000 a quarter, USD 320,000 a year.
- **Trap:** a futures contract retains notional value at expiry; an out-of-the-money option or warrant goes to zero. And the notional is never exchanged in an interest-rate swap.

Chapter 3 — Futures Strategies

Who trades futures, how orders behave, which spread isolates which relationship, and how to size a hedge. This is the most calculation-heavy chapter after Options.

Learning Objectives

The candidate should be able to:

- Understand and explain the participants and their roles in the futures market.
- Explain the various types of futures orders, futures trades and trading strategies.
- Describe and explain the different types of spread trades.
- Explain how to identify and manage financial risks using futures.
- Describe and demonstrate how to develop and manage hedging programmes for various cash positions.

- Describe and demonstrate how to develop and manage hedging for short-term and long-term interest rate risks.
- Explain how to manage a hedge programme for equity risk exposures.
- Describe and demonstrate arbitrage strategies using short-term and long-term interest rate futures.

3.1 Market Participants

The main participants are:

- **a.** Hedgers
- **b.** Speculators
- **c.** Market makers
- **d.** Proprietary trading firms
- **e.** Arbitrageurs
- **f.** Portfolio managers
- **g.** Hedge funds
- **h.** Central banks, sovereign funds and regulators

As volatility rises, participants with different motivations turn to derivatives to hedge or to maximise return. The same firm can act as speculator, hedger and arbitrageur at once, and knowing which group is driving the market helps decide whether to follow or oppose the direction. Growth in hedge funds and alternative assets has driven innovative derivative strategies, and growth in wealth management has pushed institutions to offer structured products built from futures and other derivatives.

3.1.1 Hedgers

Hedgers hold a position in the underlying asset or commodity and use futures to reduce or limit the risk of an adverse price change. Producers such as farmers sell futures on the crop they grow to hedge a price fall, which makes long-term planning easier. Industrial users buy futures to fix input costs — a food processor fixing corn or wheat, an airline hedging fuel, a jeweller hedging gold and silver — so they can plan on a fixed cost for core supplies and stabilise the price passed to customers.

3.1.2 Speculators (Individual Traders)

Many speculators are individuals trading their own funds, expressing a view on a market's direction. Electronic trading has levelled the field by improving access to price and trade information; speed and ease of execution, with modern risk management, give individuals access to markets and strategies once reserved for institutions.

3.1.3 Market Makers

Market makers are trading firms contractually obliged to provide liquidity, continually quoting both bids and offers, usually in exchange for reduced trading fees. Electronic market makers now supply much of the liquidity that lets large transactions occur without moving price substantially. They profit from capturing the **spread** — the small difference between bid and offer across many transactions — or by trading related futures markets where they see price or profit opportunities.

3.1.4 Proprietary Trading Firms

Prop shops profit directly from their traders' activity. They supply education and capital so traders can execute many trades a day, gaining more leverage than they would on their own account and access to research and strategies developed by larger institutions.

Some large banks retain proprietary units. The most famous loss is **Barings in 1995**, where a single proprietary trader's excessive risk-taking on **Nikkei 225 futures** accumulated losses **twice the bank's capital**. Poor controls and supervision led several banks to close proprietary units, but losses from distressed debt, CDOs, CDS and **Delta One** instruments continued to be reported well after the crisis.

> **Delta One.** Desks provide OTC derivative structures pegged one-to-one against an underlying, usually an index. The bank trades the underlying and related derivatives — sometimes swaps with other banks — seeking arbitrage profit or profit from holdings moving against the direction of the structure sold. Trades are typically very large because margins are extremely fine, and because most are OTC they are hard to monitor or regulate.

Financial institutions also hedge their sale of derivatives to institutional and private clients through futures and options, and ETF and ETN issuers manage daily offsetting futures and options positions against their long and short baskets through the proprietary desk.

3.1.5 Arbitrageurs

Arbitrageurs seek riskless profit from disequilibrium between cash and futures markets. They take no directional bet, instead exploiting price differences between related markets or markets with a fixed relationship — the gap between implied and market value. They rely on computer systems and models, because windows close very fast: computers spot the opportunity and arbitrage it away. Arbitrageurs improve liquidity and keep the pricing relationship between futures and the underlying consistent, and they usually work for market makers and proprietary firms.

3.1.6 Portfolio Managers

A portfolio or investment manager invests or hedges the assets of a mutual fund, ETF or closed-end fund, implementing the strategy and managing day-to-day trading. Futures are often used to raise or lower overall market exposure **without disrupting a portfolio balance that took significant effort to build**.

3.1.7 Hedge Funds

The deep, liquid futures market lets hedge funds execute large transactions and adjust exposure without changing security holdings. A hedge fund is a managed portfolio using advanced strategies to maximise return, absolutely or against a benchmark. The name is historical: early funds shorted the market to offset bear-market risk. On occasion hedge funds have exploited market weakness by shorting futures collectively, adding sharp downward pressure to both futures and spot.

3.1.8 Sovereign Funds, Central Banks and Regulators

Collective selling by speculators and hedge funds has sometimes forced governments to intervene in spot and futures markets to counterbalance speculative forces and stabilise the market. In severe cases, tightening liquidity, capital flows and trading activity has been used to break the momentum in both markets.

During the **1997 Asian Currency Crisis**, several Asian currencies and equity markets were driven down under persistent heavy selling, but the speculation was broken when some governments intervened in spot and futures markets and introduced **capital and currency exchange controls**, locking speculators into short currency positions they could not unwind while incurring interest cost.

Day to day, many governments manage their currency within a trading band and participate in spot and futures markets to influence its direction.

3.2 Types of Orders on the SGX

Most order types are common across futures and stock exchanges.

Order type	Behaviour
Limit order	Carries a specified price and quantity — the price the trader will pay, or wants
Market-to-limit (MTL) order	Has a validity time and can be entered as Fill-and-Kill (FAK), Fill-or-Kill (FO
Market order	Entered while orders are continuously matched, it matches at the best possible
Stop order (limit, MTL or market)	Hidden until its trigger is met, then converted to a limit, MTL or market order
Market-if-touched (MIT) order	An order to buy or sell a contract **below** or **above** the market, held until the t
Session state order (SSO)	Triggers entry of an order into the book only when the market transitions into

3.2.1 Other Types of Orders — Packs and Bundles

Packs and bundles were introduced by **CME Group** for Eurodollar contracts; NYSE Euronext LIFFE offers them for STIR Euribor and Sterling futures. They buy or sell a series of futures representing a segment along the yield curve, creating or liquidating positions along the curve and gaining exposure to longer-term rates **without legging risk** — the risk of failing to complete one leg. The advantage is execution at a single price: no multiple orders per contract, lower overall cost, and no unfilled legs.

- A **pack** buys a predefined number of contracts in **four consecutive delivery months**. Eurodollar packs are the simultaneous purchase of an equally weighted, consecutive series of four Eurodollar futures, executed in a single transaction so partial fills cannot occur.
- A **bundle** buys a predefined number of contracts in **each consecutive quarterly month for two or more years**, across the standard **2, 3 or 5-year** periods. There are **no 1-year bundles** because those are effectively packs. The first contract is typically the first quarterly contract in the strip, but a bundle can start at any quarterly contract.

Packs and bundles are quoted on the **average net change** in value of all constituent Eurodollar contracts since the prior settlement, in increments of **¼ of 1 basis point (0.01%)**.

3.3 Types of Trades and Strategies

- **a. Outright trades.** Buy or sell a single futures contract, wagering the underlying rises or falls. Familiar and easy to understand.
- **b. Hedging.** Sell a futures contract to offset a cash position. A holder of a large US equity portfolio who does not want to sell for tax reasons but fears a decline can sell S&P 500 index futures.
- **c. Basis trades.** An arbitrage strategy taking opposing long and short positions in two securities to profit from convergence. A trader could be long 10-year US Treasury **bond futures** and short physical 10-year Treasury **bonds**, expecting undervalued futures to appreciate relative to the overpriced bond. Making the trade worthwhile requires substantial **leverage** to enlarge the positions.
- **d. Spread trades.** See 3.4.

3.4 Spread Trades

A spread normally needs less margin than two unrelated outright positions but is not risk-free: correlation can break, liquidity can vanish, and one leg may fill without the other. Types:

- **Intra-commodity / intra-delivery** — same commodity and market, different delivery months.
- **Inter-commodity** — related commodities, e.g. crude against refined products.
- **Intra-market** — related contracts on the same exchange.
- **Inter-market** — similar contracts on different exchanges.
- **Calendar / horizontal** — same underlying, different months.

For a calendar spread, long the near month and short the deferred is generally a bullish view on the near-versus-deferred relationship; short near and long deferred is bearish. Profit depends on the **change in the spread**, not simply on whether the underlying rises.

3.4.1 Calendar Spread

With the S&P E-mini index at 2,344.00 on 19 April 2020, June at 2,335 and December at 2,330, a long June / short December spread requires initial margin of **USD 70,050 + USD 69,900 = USD 139,950** — margins are not netted across the legs.

Scenario	Index move	June	December	Spread P&L
1	+4%; June +6% to 2,475; Dec +5% to 2,447	Gain	Loss	**+USD 70,000 −**
2	+2.5%; June +3% to 2,405; Dec +3.5% to 2,412	Gain	Loss	**+USD 35,000 −**
3	−4% with a circuit breaker; June −7.5% to 2,160; Dec −9% to 2,120	Loss	Gain	**−USD 87,500 +**

Read the results carefully: scenario 3 makes money on a **falling** market because the deferred leg falls further, so the spread widens. The spread, not the direction, is what is being traded.

A currency example: E-micro EUR/USD, 12,500 EUR per contract, tick 0.0001 USD. Selling the near leg at 1.1810 and buying the far at 1.1866, then closing at 1.1750 and 1.1828, gives **+USD 7,500 and −USD 4,750, net +USD 2,750**.

3.4.2 Butterfly Spread

Four legs across three maturities with weights **+1, −2, +1**. It targets curvature or relative richness and is more market-neutral than an outright.

3.4.2.1 Buying a Butterfly **Gold butterfly.** On 6 January 2020 the April, June and August contracts stood at 1,176.3, 1,179.0 and 1,181.7; by 21 April they were 1,287.4, 1,289.1 and 1,292.4. The two spreads widened to **+1.00 and +0.60**, a net **+1.60**. With 100 oz per contract and 100 contracts:

$$100 \times 1.60 \times 100 = \text{USD } 16{,}000$$

The same +1.60 recurred on 24 April after the French election.

3.4.2.2 Selling a Butterfly **Eurodollar sell-butterfly** on 8 February 2020: short 1 Jun-20 at 98.8000, closing at 98.7350; long 2 Dec-20 at 98.5325, closing at 98.5550; short 1 Jun-21 at 98.3250, closing at 98.3800. The results are **+0.0875, −0.0325** on two contracts, and **−0.0550**, giving a net **+0.0550 = 5.5 ticks**. One basis point is one tick worth **USD 25**:

$$5.5 \times 25 = \text{USD } 137.50$$

3.4.2.3 The Double Butterfly A butterfly is bought and another sold in staggered fashion, with the second and third legs of the first butterfly being the first and second legs of the second.

Stage	Near	Mid	Mid	Distant
Buy first butterfly	Buy 1	Sell 2	—	Buy 1
Sell second butterfly, matched to the two mid-term sells	—	—	Buy 2	Sell 1
Overall structure	**Buy 1**	**Sell 3**	**Buy 3**	**Sell 1**

3.4.3 Condor Spread

A condor also combines a bear and a bull spread, but the four expiry dates are **all different**, with equally distributed delivery months and a ratio of **+1 : −1 : −1 : +1**. Buying a condor means buying the nearest month, selling the next, selling the next, and buying the farthest. It typically has a wider but flatter profit zone than a butterfly.

A Eurodollar condor on 8 February 2017: May 98.8450, June 98.8000, September 98.6675, Mar-21 98.4350. By 20 April they stood at 98.7875, 98.7300, 98.6250 and 98.4650. The near-to-mid move is **+1.25 ticks**, the mid-to-distant **+7.25 ticks**, net **+8.5 ticks**:

$$8.5 \times 25 = \text{USD } 212.50$$

The tick value is derived from

$$0.01 \times 0.01 \times 1{,}000{,}000 \times \frac{90}{360} = \text{USD } 25$$

3.4.4 Other Spread Trades

3.4.4.1 TED Spread The **TED spread** compares Treasury bill and Eurodollar (LIBOR) pricing. It is a **credit and liquidity** indicator, not a pure interest-rate measure: as the spread widens, default risk is perceived to rise and investors prefer safer assets; as it narrows, perceived default risk falls. Three-month T-bill futures were delisted from the CME, so the spread is now measured between the T-bill rate and 3-month USD LIBOR.

3.4.4.2 Inter-Product or Inter-Commodity Spread Takes advantage of the value differential between two or more related commodities or financial products influenced by similar factors. A trader may look at the spread between two financial products — US T-bills and Eurodollar — as rates move, or between two related commodities such as **gasoline and heating oil** as crude moves. Rather than taking exposure to one product, the trader exploits the greater sensitivity and reaction of one relative to the other.

3.5 Hedging and Portfolio Management

Forwards and futures were created to hedge the future delivery price of a commodity. The same principle hedges or partially hedges a fall in portfolio value.

3.5.1 Identifying and Measuring Risk

Hedging reduces risk: an adverse move is contained or profit protected, but a favourable move has capped profit. Hedging does not eliminate price risk — it converts price risk into **basis risk** between the underlying and the hedging instrument, confining the final price to a determinable range.

In portfolio management, hedging is used to:

- **a.** smooth cash flows;
- **b.** simplify financial planning;
- **c.** reduce working capital requirements;
- **d.** allow more efficient product pricing; and
- **e.** manage inventory more efficiently.

Before hedging, evaluate whether the likelihood and impact justify the cost and whether the hedge is dependable. Some subjectivity remains. Six questions structure the decision:

Factor	Question
1. Probability of negative change	What is the probability of the price or rate change? What is the likelihood o
2. Probable size of negative change	What is the likely size of the change? What was the past record?
3. Overall impact of negative change	What is the risk value of not hedging — small or large?
4. Transaction costs	What are the brokerage, financing, initial and variation margin costs? How d
5. Percentage of value to be hedged	What percentage of principal does the transaction cost represent?
6. Assumed basis risk	What is the basis risk for the hedge?

The overall assessment is subjective and not fully quantifiable, but traders estimate a **risk value** and a **hedging cost**: if risk value exceeds hedging cost, hedge.

Worked risk-value decision A treasury exposure has a **45%** probability of an adverse change of **10–11.5%**, giving a risk value of **5.18%** of principal ($0.45 \times 11.5\%$). The hedge costs, on a principal of **USD 1 million**:

Cost	Calculation	Amount
Brokerage	Given	USD 15.00
Initial margin	$1{,}500 \times 12\% \times 3/12$	USD 45.00
Additional margin	45 ticks	USD 33.75
Total hedging cost		**USD 93.75**

As a percentage of principal:

$$\frac{93.75}{1{,}000{,}000} = 0.009375\%$$

Adding an assumed basis risk of **3.0%**, total cost is **3.009375%**, which is **below** the **5.18%** risk value, so the decision is to **hedge**.

3.5.2 Developing an Effective Hedge Program

- **a. Risk exposure.** Evaluate exposure in assets and liabilities by group, maturity or profit centre.
- **b. Objective.** Set the hedge objective as an acceptable risk level or tolerance, measured by beta, volatility, value at risk or another risk unit.
- **c. Hedgeability.** Determine it by price correlation analysis, using **cross hedges** where necessary — hedging a security with a positively correlated instrument with similar price movement.
- **d. Hedge vehicle.** Choose an instrument with a high correlation to the target security.
- **e. Target.** Determine the rate or price to lock in.

3.5.3 Determining the Hedge Instrument

Find an instrument highly correlated with the target. Where the target differs from the futures contract, evaluate the yield relationship and correlation between the two, measured by regression. Examples:

- **a.** A long-term government bond is better hedged with **30-year T-bond futures** than 3-month Eurodollar futures, because long-bond rates correlate more highly with 30-year T-bond rates.
- **b.** T-bills are better hedged with **matching T-bill futures** than Eurodollar futures: in distress, T-bill futures prices rise faster than Eurodollar futures and the Eurodollar hedge becomes ineffective.
- **c.** US T-bond futures hedge the **interest-rate** risk of US corporate bond portfolios; **credit quality and sector risk are not hedged**.
- **d.** Economic factors hit sectors differently. In distress, funds switch from corporate to Treasury bonds, widening credit spreads and pushing corporate prices down relative to governments. Basis risk on a Treasury-futures hedge of corporate bonds is therefore greater, and the hedge less effective.

Futures and exchange-traded options carry four limitations as hedges:

- **i.** round numbers of contracts;
- **ii.** margin maintenance;
- **iii.** time horizon and expiry dates; and
- **iv.** no exact hedge instruments.

3.5.4 Determining the Target Rate

An investor may want to lock in a currency rate, interest rate or asset price. A **minimum variance hedge** held to the futures delivery date locks in the futures rate. If the hedge is lifted **before** delivery, the hedger cannot be assured of a spot or futures rate, because the basis changes.

$$\text{Target rate for the hedge} = \text{futures rate} + \text{target-rate basis}$$

For a long cash exposure hedged with a short futures position:

$$V_t = S_t + (F - F_t) = F + (S_t - F_t) = \text{initial futures price} + \text{ending basis}$$

3.5.5 Current and Anticipated Positions

3.5.5.1 For a current cash position **a. Weak form cash hedge.** The portfolio's time horizon is not known, so the goal is to minimise variance in the rate of return for a specified holding period.

b. Strong form cash hedge. The horizon is finite or known; the goal is to protect the portfolio by minimising variance in expected total return for a given investment period. This is **immunisation** — matching asset and liability durations. A cash and futures portfolio is created and maintained with the same interest-rate sensitivity as a **zero-coupon bond whose initial maturity equals the investment period**. The zero-coupon bond's value and rate sensitivity are tracked continuously: whenever the cash portfolio's sensitivity falls below the zero-coupon bond's, futures are **bought** to raise it; whenever it exceeds the zero-coupon bond's, futures are **sold** to reduce it.

3.5.5.2 For an anticipated cash position **a. Weak form anticipated hedge.** The goal is to minimise variance in the rate of return on an asset for a specified holding period, based on cash flows to be received at an **unknown** date. Futures are bought with the same interest-rate sensitivity as the asset to be acquired. Uncertain timing reduces effectiveness, though futures can still narrow the range of outcomes.

b. Strong form anticipated hedge. A known amount of cash will be received on a **certain** date; the goal is to minimise variance of the acquisition price. The hedge requires acquiring future delivery of the anticipated dollars of bonds, and because margin calls on the long futures position must be financed, the appropriate hedge ratio is **not an exact dollar match**.

3.5.6 Hedge Ratio

The hedge ratio is the expected movement in the value of the cash instrument given a movement in the futures contract; formally, the ratio of futures to spot that minimises or eliminates risk. Equal dollar exposure does not create an optimum or **delta-neutral** hedge.

$$h = \frac{\Delta S}{\Delta F}$$

where, for the hedged position,

$$V = S + h \times F$$

and

$$V$$

is the value of the hedged position,

$$S$$

the security price and

$$F$$

the futures price. For a complete or delta-neutral hedge,

$$\Delta V = 0$$

:

$$h = -\frac{\Delta S}{\Delta F}$$

Number of contracts:

$$\text{contracts} = \frac{-h \times \text{exposure value}}{\text{contract size}}$$

Worked example. A **USD 100 million, 180-day** loan. A 1% rate move on a 180-day basis changes value by:

$$0.01 \times \frac{180}{360} \times 1{,}000{,}000 = \text{USD } 5{,}000$$

per million, and a 90-day Eurodollar contract moves by 100bp × USD 25 = **USD 2,500**. So:

$$h = \frac{5{,}000}{2{,}500} = 2$$

and the hedge needs:

$$2 \times \frac{100\text{m}}{1\text{m}} = \mathbf{200} \text{ contracts}$$

Sell **200** contracts for the 180-day exposure. A 90-day exposure would need 100. Two further points: holding a stack of 90-day contracts against a 180-day exposure creates **basis risk at the roll**, and margin calls arise before the hedge pays off. The ratio must be recalculated when the stack is rolled.

Strips and Stacks A **strip** uses sequential futures delivery months matched to the exposure periods — three-month, six-month, nine-month — so each resets against its own contract. A **stack** concentrates the whole exposure in one or deferred contracts and rolls them; it is cheaper and more liquid but carries roll and basis risk. Choose between them by **time horizon**: an interpolative hedge uses a contract expiring beyond the exposure and interpolates; an extrapolative hedge uses a contract expiring before it.

3.5.7 Strips and Stacks

See the strip and stack definitions above: a strip matches sequential delivery months to the exposure periods, a stack concentrates the exposure in one contract or a deferred month and rolls it. Strips avoid roll risk but carry more contracts and lower liquidity per month; stacks are cheaper and more liquid but expose the hedger to basis at each roll.

3.5.8 Hedging Long-Term Interest Rate Risk

Compare **dollar interest-rate sensitivity**, not face value:

$$\text{hedge ratio} = \frac{\text{PVBP of the hedged security}}{\text{PVBP of the cheapest-to-deliver bond}} \times \text{conversion factor of the CTD}$$

where PVBP is the change in price for a one basis point change in yield.

Worked example. A fixed-income manager, Alice Tan, hedges a **USD 20 million** bond position with Treasury futures. The CTD conversion factor is **0.91**, the CTD's PVBP at the settlement date is **0.6895**, and the PVBP of the bond to be hedged is **0.5954**. Assuming a fixed yield spread between the two bonds:

$$\text{hedge ratio} = \frac{0.5954}{0.6895} \times 0.91 = 0.7858071 \approx 0.79$$

$$\text{contracts} = -0.7858071 \times \frac{20{,}000{,}000}{100{,}000} = -157.16$$

so **sell 157 contracts**.

3.5.9 Hedging Equity Risks

An equity portfolio carries two risks: **systemic or market risk** and **specific or non-market risk**. Equity index futures hedge only market risk. A manager offsets market exposure by selling futures when long in the market — a **short hedge** — or buying futures when short — a **long hedge**.

Two factors decide the number of contracts:

- **i. Beta.** Measures portfolio risk or volatility relative to the index. The manager must decide whether the portfolio should be more or less volatile than the index, usually through regression on the historical betas of the constituent stocks.
- **ii. Modified Portfolio Value (MPV).** The value of the actual portfolio multiplied by the **weighted beta** — the sum of each stock's beta multiplied by its percentage share of the portfolio. This forms the base hedge.

$$N = \frac{V_P}{F \times T} \times \beta$$

where

$$N$$

is the number of contracts,

$$V_P$$

the current portfolio value,

$$F$$

the current futures quote,

$$T$$

the value per tick and

$$\beta$$

the portfolio beta. The formula assumes the futures maturity is close to the hedge maturity.

Worked example. John Lim hedges an equity portfolio over three months with 3-month futures. The S&P index is **1,000** in September, December S&P 500 futures are **1,010**, the portfolio is worth **USD 5,050,000**, its beta is **1.50**, and each contract is worth **USD 250**:

$$N = \frac{5{,}050{,}000}{250 \times 1{,}010} \times 1.50 = 20.0 \times 1.50 = 30.0$$

so **sell 30 December contracts**.

December leg. The index falls to **900**, futures to **902**, and the portfolio falls 15% to **USD 4,292,500**. The futures gain:

$$(1{,}010 - 902) \times 30 \times 250 = \text{USD } 810{,}000$$

Total fund value:

$$4{,}292{,}500 + 810{,}000 = \text{USD } 5{,}102{,}500$$

3.5.10 Structuring the Hedge

Contract month, timing and maturity sensitivity must be chosen against the exposure before the hedge is executed. Match the futures maturity as closely as the exposure allows, keep margin headroom for variation calls, and decide the lift date in advance.

3.5.11 Managing the Hedge

Timing, contract month and maturity sensitivity must be adjusted as the hedge progresses. A hedge with **90% effectiveness leaves 10% residual standard deviation**: with a 10bp standard deviation on the basis, the hedged rate lands within ±10bp about **67%** of the time. Lift-date basis and cross-hedge parameter error are the main sources of that residual. Adjust the contract month as the exposure rolls, re-run the ratio when basis moves, and decide in advance at what level the hedge is lifted.

3.5.12 Further Examples of Hedging

Case	Setup	R
Yankee Trading Co.	CAD/USD spot 0.9125, April futures 0.9235; on 1 April spot 0.9485	C
Fixed deposit hedge	5 January, USD 25m; Eurodollar March 97.75 (2.25%); 5 March 98.16 (1.84%)	D
LMN Corp.	27 July, Eurodollar September 98.62 (+1.38%); 10 September 99.40 (−0.60%)	D
Floating-rate note	USD 10m, 3-month LIBOR + 1%; LIBOR 1.85% rising to 2.25%; Eurodollar 98.15 to 97.75	E
Nikkei hedge	JPY 530m, beta 1, June-18 futures 21,455, contract JPY 500	**4**
130/30	Long 130%, short 30%	N

3.6 Portfolio Management Techniques

Futures implement asset allocation, tactical rebalancing, cash equitisation, duration management, synthetic equity and bond exposure, market overlays and loss control. They are cheaper, more liquid and more margin-efficient than trading every cash security, but add basis, roll, leverage, margin and tracking risk. Separate market risk from security-specific risk: an index hedge will not perfectly protect a concentrated portfolio.

3.6.1 Examples of Portfolio Management Applications

a. Asset allocation switch. A USD 50 million portfolio moving from 50/50 to 40/60 equity/bond is implemented with bond futures, avoiding the cost and disruption of selling 50% of the equity book.

b. Synthetic money market. A corporate bond portfolio is converted to synthetic short-term exposure by selling T-bond futures, without liquidating the credit positions.

c. Jessie Goh. A USD 50 million portfolio with 10% into a short-term instrument at an 8% coupon. She buys July at 101 with 2.00 accrued, so the price is 103.00. In September the futures are **108 13/32 = 108.406**, conversion factor 1.0, accrued 3.33, giving proceeds of **111.736** and a gain of **8.736**, or **8.48%**. The hedge ratio:

$$N = \frac{5{,}000{,}000}{108{,}406.25} = 46.12 \rightarrow \text{sell } 46 \text{ contracts}$$

Loss control also uses futures to defer loss realisation across reporting periods.

3.7 Arbitrage

Arbitrage strategies include cash-and-carry, reverse cash-and-carry, bond stripping and reconstitution, futures-versus-IRS, FRA-versus-futures and option parity.

3.7.1 Arbitrage Strategies

3.7.1.1 Cash versus Futures Fifty days at 1.00/1.0625, 91 days, and 141 days at 1.25/1.3125 give

$$r_2 = 1.351\%$$

. The theoretical futures price is **98.65** against a market **98.72**:

$$(9{,}872 - 9{,}865) \times 25 = \text{USD } 175$$

3.7.1.2 FRA versus Futures The interbank FRA trades at a 1–3bp spread against a 1bp futures spread, and **value-date mismatch** leaves residual basis and fixing risk. Avoid calendar events and data releases falling between the two dates.

3.7.1.3 Futures versus Interest Rate Swaps IMM-dated, quarterly-fixing swaps align with futures dates. A speculative variant takes only the partial leg to express a curve view.

3.7.2 Arbitrage using Options

Put-call parity violations are arbitraged with conversion and reversal strategies, which Chapter 4 covers.

Arbitrage is risk-free only in the idealised model. Value dates, fixing dates, margin, liquidity, bid-ask spread, funding, delivery options, basis and counterparty constraints turn it into **relative-value trading**.

3.8 Summary

Know why each participant trades — hedge, speculate, make markets or arbitrage — what each spread isolates, how to compute the hedge ratio, and how margin and mark-to-market change cash needs. A hedge reduces a targeted risk; it does not remove all risk.

Exam cues

- **Hedge ratio for a rate exposure** is the ratio of dollar sensitivities, then scaled by exposure over contract size. The **180-day** USD 100m case needs **200** contracts; a 90-day exposure needs 100.
- **Bond hedge:** PVBP ratio × conversion factor. **0.5954 / 0.6895 × 0.91 = 0.79**, and 0.79 × 20m/100,000 = **157 contracts to sell**.
- **Equity hedge:**

  $$N = V_P/(F \times T) \times \beta$$

  . The John Lim case is 5,050,000/(250 × 1,010) × 1.5 = **sell 30**; with the index at 900 and futures at 902 the hedge recovers USD 810,000.
- **Butterfly weights +1, −2, +1 through three maturities; condor +1, −1, −1, +1 through four.** A double butterfly nets **buy 1 / sell 3 / buy 3 / sell 1**.
- **Eurodollar tick = USD 25**; one basis point is one tick. The sell-butterfly earned 5.5 ticks = **USD 137.50**; the condor 8.5 ticks = **USD 212.50**.
- **Order types:** MTL leaves the unfilled balance at the last execution price; a market order in a non-continuous state takes the **Equilibrium Price**; MIT sells are placed **above** the market while stop sells are **below**; SSOs **cannot be GTC** and vanish at end of day.
- **Packs are four consecutive months; bundles are 2, 3 or 5 years** — there is no 1-year bundle, because that is a pack. Quoted in **¼ basis point** increments.
- **TED spread** is a credit and liquidity indicator, not a pure rate spread. It narrowed in measurement after 3-month T-bill futures were delisted.
- **Trap:** equal dollar exposure does not give a delta-neutral hedge, and an equity index hedge removes **market** risk only — specific risk and credit spread remain.

Chapter 4 — Options

Options separate rights from obligations, which is why they can be priced, hedged and combined far more finely than futures. This chapter builds from payoff arithmetic through the Greeks to the strategy matrix the examination tests.

Learning Objectives

The candidate should be able to:

- Understand the basic concepts of options and how they are traded.
- Explain the features of call and put options.
- Describe the major factors affecting the price sensitivity of options.
- Explain the rights, obligations, benefits and risks of using options.
- Calculate breakeven, profit and loss from an option contract.
- Explain how traders use options strategies.
- Explain neutral, bullish and bearish strategies based on the investor's outlook.
- Explain how to use options to hedge an underlying position.
- Describe options on equities, bonds, interest rates, currencies, futures and swaps.

4.1 What is an Option?

An option is a contract between the **option writer** (seller/issuer) and the **option holder** (buyer) giving the holder the right to buy from, or sell to, the writer at a contracted strike price within a specific period. There are two types: **calls** and **puts**.

The exchange-listed market dates from **1973**, when the Chicago Board Options Exchange listed calls on **16 stocks**; before that, options were OTC only. Singapore's options market remains largely OTC, which makes counterparty scrutiny important.

Call option:

- **a. Buyer.** Has the right but not the obligation to buy the underlying at the strike within the period. On exercise the holder pays the strike and takes possession.
- **b. Writer.** Receives the premium and must deliver the underlying if assigned.

Put option:

- **a. Buyer.** Has the right but not the obligation to sell the underlying at the strike.
- **b. Writer.** Receives the premium and must take delivery at the strike if assigned.

Exotic features include compound options, and underlyings extend to real estate, weather, volatility and ETFs.

4.2 Style of Option

4.2.1 Style of Option

European options are exercisable only at expiry; **American** options at any time up to expiry. OTC contracts may use **Bermudan** or event-dependent exercise. US listed equity options conventionally expire on the **third Friday** of the month, moving to the Thursday if that Friday is a holiday.

4.2.2 Strike Price

The contractual price at which the underlying is bought or sold on exercise.

4.2.3 Expiration Date

The last date on which the option may be exercised. Listed series follow monthly cycles coded **JAJO** (January, April, July, October), **FMAN** (February, May, August, November) and **MJSD** (March, June, September, December).

4.2.4 Class and Series

A **class** is all options of the same type on the same underlying; a **series** is all options of the same class with the same strike and expiry.

4.3 Option Price

$$\text{option price (premium)} = \text{intrinsic value} + \text{time value}$$

4.3.1 Intrinsic Value

For a call, exercise has value when the market price exceeds the strike; for a put, when the strike exceeds the market price.

$$IV_{\text{call}} = \max(0,\ S - X) \qquad IV_{\text{put}} = \max(0,\ X - S)$$

Moneyness	Call	Put
In-the-money (ITM)	$S > X$, positive intrinsic value	$X > S$, positive intrinsic value
At-the-money (ATM)	$S = X$, zero intrinsic value	$X = S$, zero intrinsic value
Out-of-the-money (OTM)	$S < X$, zero intrinsic value	$X < S$, zero intrinsic value

Call example. Strike

$$X = 50$$

, premium

$$P = 6$$

, underlying

$$S = 54$$

. Intrinsic value is **4** and time value **2**. If the underlying is 48 at expiry, intrinsic value is 0.

Put example. Strike

$$X = 50$$

, premium

$$P = 9$$

, underlying

$$S = 44$$

. Intrinsic value is **6** and time value **3**. If the underlying is 53 at expiry, intrinsic value is 0.

4.3.2 Time Value of Options

Time value is generally positive and falls to zero at expiry, so at expiration the option is worth exactly its intrinsic value:

$$\text{option price} = \text{intrinsic value as time value} \to 0$$

The longer the time to expiry, the greater the time value. An option that looks worthless today — ATM or OTM, zero intrinsic value — may be in the money by expiry.

Criterion	Call buyer	Call seller	Put buy

4.3.3 Comparison of Call versus Put

Criterion	Call buyer	Call seller	Put buy
Obligation	Pays the premium	Delivers the underlying if assigned	Pays the
Right	May buy at the strike	—	May sell
Maximum gain	$U_{Mkt} - U_{Strike} -$ premium, theoretically unlimited	The premium	$U_{Strike} -$
Maximum loss	The premium	Theoretically unlimited	The pre
Counterparty risk	Bears the writer's	Bears the buyer's	Bears th
Opportunity cost	—	Forgone upside on a covered call	—

4.4 Factors Influencing Premiums

Change, all else constant	Call value	Put value
Underlying price rises	Rises	Falls
Strike rises	Falls	Rises
Time to expiry rises	Generally rises	Generally rises
Volatility rises	Rises	Rises
Risk-free rate rises	Rises	Falls
Dividends rise	Falls	Rises

4.4.1 Underlying Price

A call gains as the underlying rises; a put loses.

4.4.2 Strike Price

A higher strike makes a call cheaper and a put dearer.

4.4.3 Time to Expiration

More time means more chance of finishing in the money, so both calls and puts carry more time value.

4.4.4 Volatility of the Underlying

Higher volatility widens the distribution of outcomes and raises the value of both calls and puts, because the holder's loss is floored at the premium while the upside is open.

4.4.5 Interest Rates

The interest forgone on the cash needed to buy the underlying — the opportunity cost, measured against a risk-free alternative such as Treasury bills — is embedded in the option. The higher the rate, the greater that opportunity cost, so there is a **positive relationship between interest rates and call value**; puts move inversely.

4.4.6 Expected Dividends

For equity options, a dividend reduces the share price when it goes ex-dividend, which lowers call value and raises put value.

4.5 Payoff Profiles

A payoff profile shows maximum gain, maximum loss and breakeven as a function of the underlying at expiry.

4.5.1 Buying a Call

An investor bought a call a month ago with premium **USD 4**, strike **USD 100**, and the share is now **USD 110**.

Payoff to the call buyer:

$$C_T = S_T - X \text{ if } S_T > X \text{ (ITM)}; \quad C_T = 0 \text{ if } S_T \leq X$$

Item	Result
i. Payoff and profit	Payoff is **USD 10** $(110 - 100)$; profit after premium is **USD 6**
ii. Breakeven	The share must rise above the strike by the premium: **USD 104**
iii. Maximum gain	Theoretically unlimited
iv. Maximum loss	The premium paid, **USD 4**; if $S_T < X$ exercise does not occur and the option expires worthless
v. Obligations	Apart from paying the premium, none unless the buyer exercises, and then to pay the strike

4.5.2 Selling a Call

An investor sells a call with premium **USD 4**, strike **USD 100**, share at **USD 90**.

Payoff to the writer:

$$C_T = -(S_T - X) \text{ if } S_T > X; \quad C_T = 0 \text{ if } S_T \leq X$$

Item	Result
i. Payoff and profit	Below USD 100 there is no reason to exercise, so payoff is zero and profit is the **USD 4 premiu**
ii. Breakeven	**USD 104** — identical to the buyer's
iii. Maximum gain	The **USD 4** premium, when the share is at or below the strike
iv. Maximum loss	Theoretically unlimited
v. Obligations	Must deliver if assigned

4.5.3 Buying a Put

An investor buys a put with premium **USD 4**, strike **USD 130**, share at **USD 120**.

Item	Result
i. Payoff and profit	Payoff **USD 10** $(130 - 120)$; profit **USD 6**
ii. Breakeven	**USD 126** $(130 - 4)$
iii. Maximum gain	**USD 126** at an underlying of zero, being the strike less the premium
iv. Maximum loss	The premium, **USD 4**
v. Obligation	None except on exercise

4.5.4 Selling a Put

An investor sells a put with premium **USD 4**, strike **USD 130**, underlying at **USD 140** initially and **USD 120** at expiry.

Payoff to the writer is the mirror image of the holder:

$$P_T = -(X - S_T) \text{ if } S_T < X; \quad P_T = 0 \text{ if } S_T \geq X$$

Item	Result
i. Payoff and profit	Zero above USD 130, with the **USD 4** premium as maximum profit; below the strike the profit e
ii. Breakeven	**USD 126**
iii. Maximum gain	The **USD 4** premium

Item	Result
iv. Maximum loss	**USD 126**, when the underlying falls to zero
v. Obligation	Must take delivery at the strike if assigned

4.6 Put-Call Parity

4.6.1 Definition

For European options on the same underlying, strike and expiry, with consistent units and dividends correctly treated:

$$C + \mathrm{PV}(X) = P + S$$

The relationship assumes no arbitrage, a frictionless market, known dividends, the same exercise style and no early exercise.

4.6.2 Arbitrage

A violation is traded as a **conversion** or **reversal**. If

$$C + \mathrm{PV}(X) > P + S$$

, sell the call, buy the put and buy the stock. If the left side is cheaper, reverse the trade. Funding, dividends, exercise style, transaction costs and short-sale constraints determine whether the arbitrage is real.

4.6.3 Synthetic Securities

Synthetic position	Built from
Long stock	Long call + short put at the same strike (plus the parity bond)
Short stock	Long put + short call
Long call	Long underlying + long put
Long put	Short underlying + long call
Short call	Short underlying + short put
Short put	Long underlying + short call

Each synthetic reproduces the cash flows of the position on the left, so a trader can choose whichever route is cheaper after funding and transaction cost.

4.7 Valuation

4.7.1 Black-Scholes

The model takes six inputs — spot, strike, time to expiry, interest rate, dividends and volatility — and returns a theoretical premium. **Binomial trees** handle early exercise and American features. The examination does **not** require model calculations, but does require explanation.

4.7.2 Volatility

Volatility is the annualised standard deviation of daily price changes. **Historical** volatility is measured from past prices; **implied** volatility is backed out of the observed option price and represents the market's forward-looking estimate. The two can diverge.

4.7.3 The Option Greeks

The Greeks measure sensitivity of the premium to each input.

Greek	Measures	Typical sign for a long option
Delta	Change in option price per unit change in underlying	Positive for calls, negative for puts
Gamma	Change in delta per unit change in underlying	Positive
Theta	Time decay	Negative
Vega	Sensitivity to implied volatility	Positive
Rho	Sensitivity to the interest rate	Positive for calls, negative for puts

4.7.4 Delta

$$\delta = \frac{\partial V}{\partial S}$$

A delta of **0.50** means a **USD 2** move in the underlying changes the option price by **USD 1.00**. Delta approaches **1** for deep in-the-money options near expiry and **0** for deep out-of-the-money options.

Delta hedging. A writer short **1,000 calls** with delta **0.642** loses **USD 642** on a one-unit rise; hedging requires buying **642 units** of the underlying.

4.7.5 Gamma

$$\Gamma = \frac{\partial^2 V}{\partial S^2} = \frac{\partial \delta}{\partial S}$$

Gamma is largest at-the-money and close to expiry, when delta changes fastest. A high-gamma book requires frequent rebalancing.

4.7.6 Theta

Time decay, normally negative for a long option. At-the-money options lose the most dollar value per day: a one-day theta of **−0.0250** is **−2.5 cents a day**.

4.7.7 Vega

Sensitivity to a one-percentage-point change in implied volatility; a vega of **0.090** means the premium changes by 9 cents per volatility point. Vega is larger at-the-money and for longer-dated options. Implied volatility is obtained by inverting the pricing model against the observed price.

4.7.8 Rho

Sensitivity to a one-percentage-point change in the interest rate; a rho of **0.060** means the premium changes by 6 cents per point. Call rho is positive and put rho negative, because higher rates raise the opportunity cost of holding the underlying.

4.7.9 Practical Hedging Problems

Delta hedging is dynamic and imperfect. Rebalancing costs, jumps through levels, liquidity, changing volatility, discrete monitoring and model error all remain. A market maker manages the whole Greek book, not just delta.

4.8 Exchange versus OTC Options

Exchange options	OTC options
Standardised strikes, expiries and sizes	Tailored strike, date, notional, barrier, settlement currency
Clearing house counterparty	Bilateral counterparty and documentation risk
Transparent prices and secondary liquidity	Valuation and liquidity risk; CSA collateral terms matter

4.9 Option Contract Specifications

Read the whole term sheet, not just the ticker. An SGX equity index option specification illustrates what to check:

Term	Specification
Ticker	`CS - Calls`
Contract size	One MSCI Singapore index future
Contract months	2 nearest serial months plus the quarterly cycle (Mar/Jun/Sep/Dec)
Trading hours	T session 8.15–8.30 opening (cancel/match), 8.30–17.15; T+1 18.00–18.15 opening, 18.15–02.00; last tr
Last trading day	Second-last business day of the expiring month
Tick size	0.1 index point, worth **SGD 20**
Strike intervals	**5 index points**
Exercise style	European or American, per the series
Settlement	Cash settled; in-the-money options **auto-exercised**
Price limits	Daily limit of **5%** up or down, with circuit breakers
Position limit	**10,000 contracts** net on the same side, all months combined

4.10 Options on Other Underlyings

4.10.1 Equity Index Options

Index options were listed on the **CBOE in 1983**. They are usually cash settled, and the payoff is multiplied by the contract multiplier.

Contract	Index value	Multiplier	Style
SPX (S&P 500 options)	Full index value	**USD 100**	European
XSP (Mini-SPX)	1/10th of the index	**USD 100**	European

4.10.2 Bond Options

Bond options are mostly OTC, with expiries set well before bond maturity. A bank buying a call at a strike of **USD 102** with the bond at **USD 101** gains protection against a rise in price — that is, a fall in yield.

4.10.3 Interest Rate Options

Caps, floors and collars are the interest-rate equivalents of calls, puts and combinations. CME lists options on US Treasuries and Eurodollar; ICE on Euribor; SGX on the 10-year Mini-JGB and Euroyen (TIBOR).

4.10.3.1 Options on Eurodollar Futures Options on short-rate futures are quoted on **100 − yield**. One tick is:

$$\text{USD } 1{,}000{,}000 \times \frac{1}{10{,}000} \times \frac{90}{360} = \text{USD } 25$$

A quarter tick is **USD 6.25** in the spot month and a half tick **USD 12.50** in other months.

Mid-curve example. A 1-year Eurodollar mid-curve option on the Dec-2019 futures at **96.93**, American style, tick value USD 25, expiry June 2019, is struck on a ladder from 9,637.50 to 9,750.00. Call prices run 51.50, 42.00, 33.25, 26.25, 20.25, 15.50, 12.00, 9.25, 7.25 and 5.75; put prices 4.00, 6.75, 10.50, 15.75, 22.50, 29.75, 38.75, 48.25, 58.75 and 69.75.

Hedge arithmetic. A borrower wanting to cap the rate at 2.75% buys the strike-97.250 option (that is, 100 − 2.750) at a cost of **9.25bp**, so the realised rate is:

$$2.7500\% - 0.0925\% = 2.6575\%$$

A lender buying the put at **48.25bp** achieves:

$$2.7500\% + 0.4825\% = 3.2325\%$$

4.10.3.2 Interest Rate Caps A cap is a series of call-like options on a reference rate, setting a maximum. A **SGD 200 million 3-year cap at 2.00%** on 3-month SIBOR, struck in March 2018 when SIBOR was 1.375%, pays when SIBOR fixes at **2.125%** in June 2019:

$$\frac{90}{360} \times (2.125\% - 2.00\%) \times 200{,}000{,}000 = \text{SGD } 62{,}500$$

4.10.3.3 Interest Rate Floors A floor is a series of put-like options setting a minimum rate, protecting a lender or depositor against falling rates.

4.10.3.4 Interest Rate Collars A collar buys one option and sells another, reducing or eliminating premium at the cost of capping the benefit. A borrower buys a cap and sells a floor; the sold floor funds the bought cap.

4.10.4 Currency Options

A currency option involves **two** currencies, not one price. A USD/JPY call is economically the same exposure as a JPY/USD put. Large notionals and a 24-hour market heighten monitoring and counterparty risk.

4.10.5 Options on Futures

An option on a futures contract confers the right to enter a futures position at the strike. Such options are typically more liquid than options on the cash instrument, but they introduce a **delivery problem** when cross-hedging. All SGX options are futures options, American style, with in-the-money positions auto-exercised. The **SGX Nifty 50** option covers 50 large, liquid NSE stocks representing about 70% of free float, is cash settled on the average weighted prices of the components, and benefits from the CME mutual-offset arrangement.

4.10.6 Swaptions

A **swaption** is an option to enter a swap at a fixed rate on a future date. A **payer** swaption gives the right to pay fixed — it protects against rising rates. A **receiver** swaption gives the right to receive fixed — it protects against falling rates. The terms that matter are premium, expiry, notional, the fixed rate and floating index, payment frequency and day count.

4.11 Basic Option Strategies

4.11.1 Call Buying Strategies

Calls are bought for four reasons: capital investment, taking risk exposure, extracting cash from a position, and hedging.

Hedging a short. An investor is short at **USD 12.00** and buys a call at strike **12.40** for **0.80**. Breakeven is **11.20**; maximum loss **USD 1.20**; maximum gain **USD 11.20** if the share falls to zero.

4.11.2 Call Writing Strategies

Covered call. Buy the stock at

$$S_0 = 12.00$$

, write a call at

$$X = 12.40$$

for

$$C_0 = 0.80$$

.

$$\text{Profit} = S_T - \max(0, S_T - X) - S_0 + C_0$$

Item	Calculation	Result
Breakeven	$S_0 - C_0 = 12.00 - 0.80$	**11.20**
Maximum gain	If $S_T > X$: $X - S_0 + C_0 = 12.40 - 12.00 + 0.80$	**1.20**
Maximum loss	If $S_T = 0$: $0 - 12.00 + 0.80$	**−11.20**

The call will not be exercised once the shares have fallen, so the writer keeps the premium but bears the full fall in the stock.

4.11.3 Put Buying Strategies

Protective put. Stock at **USD 12.00**, put at strike **11.60** for **0.80**. Breakeven **12.80**; maximum loss **USD 1.20**; upside retained without limit above the breakeven.

4.11.4 Put Writing Strategies

A put writer earns premium in exchange for a conditional obligation to buy the underlying. A **bull ELN** works this way: a one-month note issued at **USD 9,900** against USD 10,000 face, a vanilla yield of 5% p.a., ABC spot **USD 10** and a put struck at **USD 9**.

- If
$$S_T \geq 9$$
, the note returns USD 10,000: a gain of 1.01% for the month, or **12.82% p.a.**
- If
$$S_T < 9$$
, the note delivers **1,111 shares**. At **USD 8.80** those are worth USD 9,776.80, a loss of **1.24%**.

The uplift over the vanilla note — **7.82% p.a.** — is exactly the premium on the embedded short put.

4.12 Market Driven Strategies

4.12.1 Neutral Strategies

Used when the investor expects little movement. Premium is collected by selling options, and the risk is a break out of the range.

4.12.2 Bullish Strategies

Long call, bull call spread, bull put spread. The spread versions cap both gain and loss.

4.12.3 Bearish Strategies

Long put, bear put spread, bear call spread.

4.12.4 Creating Spreads

A spread pairs a long and a short option of the same type, differing in strike or expiry. **Debit** spreads pay premium — bull call, bear put. **Credit** spreads receive premium — bull put, bear call.

4.12.5 Straddle

A long straddle buys a call and a put at the **same strike**. With ABC at **USD 20** in September, buying the October 20 call and the October 20 put produces three outcomes at expiry:

Outcome	Result
$S_T = 20$	Both expire worthless; the loss is the whole cash outlay
$S_T > 20$	The call is in the money; profit is $S_T - 20 -$ outlay, potentially unlimited
$S_T < 20$	The put is in the money; profit is $20 - S_T -$ outlay, with a maximum payout of USD 20 at zero

The strategy is neutral on direction and **long volatility**. Maximum loss is the cash outlay, occurring precisely at the strike.

4.12.6 Strangle

A strangle buys a call and a put at **different** out-of-the-money strikes. With ABC at USD 20, buy the October 25 call and the October 15 put. It costs less than a straddle but needs a larger move. The **short** strangle does the reverse: selling the call for USD 2 and the put for USD 1 collects **USD 3**. At an underlying of 14 the put loses USD 1 while the call gains USD 2; at 26 the call loses USD 1.

4.12.7 Bull Call Spread

ABC at **USD 21**. Buy the October 20 call at **1.50** and sell the October 25 call at **0.75**, for a net debit of **0.75**.

Item	Result
Breakeven	**20.75**
Maximum profit	**4.25** (the 5-point strike gap less the 0.75 debit)

4.12.8 Bull Put Spread

ABC at **USD 24**. Buy the October 20 put at **0.75** and sell the October 25 put at **1.25**, for a net credit of **0.50**. Maximum loss is **4.50**; at an underlying of 23 the result is **−1.50**.

4.12.9 Bear Call Spread

ABC at **USD 18**. Buy the October 20 call at **1.50** and sell the October 15 call at **3.50**, for a net credit of **2.00**. Maximum loss is **3.00**.

4.12.10 Bear Put Spread

ABC at **USD 18**. Buy the October 15 put at **1.00** and sell the October 20 put at **3.00**, for a net debit of **2.00**. Maximum gain is **3.00**.

4.12.11 Other Market Driven Strategies

A three-by-three matrix of outlook against expected volatility selects the structure: the direction (bullish, neutral, bearish) fixes which legs to hold, and the volatility view fixes whether to be long or short premium.

View	Structure	Exposure
Strongly bullish	Long call, bull call spread	Rise
Strongly bearish	Long put, bear put spread	Fall
Large move, direction unknown	Long straddle or strangle	Long volatility
Range-bound, low volatility	Short strangle, iron condor	Short volatility, tail risk
Neutral at one strike	Butterfly	Limited central profit
Neutral over a wider zone	Condor	Wider zone, lower maximum

4.13 Hedging

4.13.1 Protect an Equity Portfolio

A protective put floors the downside at the strike after the premium. With ABC at USD 20, selling the USD 30 call for 1.50 is a **covered call**: the premium is earned but the upside above 30 is surrendered — the opportunity cost of the strategy.

4.13.2 Manage Currency Exposure

An importer with USD 2 million payable, USD/JPY at 100, buys a USD call / JPY put struck at 105. The exporter's mirror position is set out below.

Spot at expiry	Unhedged proceeds	Hedged with a 105 put at JPY 1 premium
100	100	**99**

Spot at expiry	Unhedged proceeds	Hedged with a 105 put at JPY 1 premium
110	110	**109**
90	90	**94**

The hedge floors the rate at 94 while retaining the upside above 105, less the JPY 1 premium.

4.13.3 Zero-Cost Options

A **zero-cost collar** buys a put and sells an out-of-the-money call so the premiums roughly offset.

NVDA costless collar. Spot **USD 156.00**. Sell 1,000 January-2020 **USD 160** calls at **USD 15.30**; buy 1,000 January-2020 **USD 155** puts at **USD 15.30**. Net outlay is zero.

- Below 155, say 125: the put pays USD 30 against a stock loss of USD 31 per share, so a residual loss of **USD 1,000** across the position.
- Between 155 and 160: neither option pays.
- Above 160: the stock is called away; upside is capped at the call strike.

The collar is not free of risk — the residual USD 1,000 is the gap between the two strikes — and it assumes no dividends and no interest over the life.

4.13.4 Summary of Options Strategies

The full strategy map crosses the market view — yield, bullish, neutral, protective, bearish — against the structure, and records whether each is a debit or credit, its maximum gain and its maximum loss. Learn the payoff of every single leg before combining them; the combination is only the sum of the legs.

4.14 Summary

Options separate rights from obligations. Price is intrinsic plus time value, and volatility, time, rates and dividends drive the premium. Learn payoff, breakeven and maximum loss for each leg before combining legs. Parity explains synthetic positions and the Greeks explain risk management.

Chapter 4 Appendix — Additional Spreads (non-examinable)

Long Call Butterfly

With strikes at **USD 10, 20 and 30**, buy one in-the-money call, sell two at-the-money calls and buy one out-of-the-money call. Maximum gain is USD 10 less the net outlay; maximum loss is the outlay.

Long Put Butterfly

Strikes at **USD 15, 25 and 35**, the mirror construction. Maximum gain is USD 10 less the outlay.

Vertical, Horizontal and Diagonal Spreads

A **vertical** spread uses the same expiry with different strikes. A **horizontal** or calendar spread uses the same strike with different expiries. A **diagonal** spread differs in both strike and expiry.

Condor Spread

	Long	Short
Calls	Buy the low strike, buy the high strike	Sell the two middle strikes
Puts	Buy the low strike, buy the high strike	Sell the two middle strikes

Ratio Hedge

A 2:1 spread: two options sold against one bought. Upside is limited while downside can become very large.

Seagull Strategy

A three-legged structure with zero net outlay. With USD/JPY at **113**, sell the 109 put, sell the 115 call and buy the 111 put. At a spot of 110 the position can be closed for a profit.

To **re-centre**, buy back the 115 call and 109 put for **−1.50%**, then sell the 112 call and 107 put for **+1.35%**, for a net **−0.15%**, while retaining the in-the-money 111 put.

Protecting or Re-centring a Short Strangle

A three-month roll and adjustment sequence: as the market moves against one leg, buy back the threatened option and sell a further out-of-the-money option to fund it, keeping the position short premium but re-centred on the new range.

Appendix Strategy Summary

The appendix closes with a grid recording, for each structure, the market view, debit or credit, maximum gain and maximum loss.

Exam cues

- **Premium = intrinsic + time value.** Call IV = max(0, S − X); put IV = max(0, X − S). Time value is zero at expiry.
- **Breakeven:** long call **X + premium**; long put **X − premium**. Buyer and writer share the same breakeven.
- **Maximum loss:** long call and long put lose the premium. A naked short call is unlimited; a short put loses down to a zero underlying, so its maximum loss is X − premium.
- **Payoff arithmetic to memorise:** call premium 4, strike 100, share 110 → payoff 10, profit 6, breakeven 104. Put premium 4, strike 130, share 120 → payoff 10, profit 6, breakeven 126, maximum gain 126.
- **Parity**
$$C + \mathrm{PV}(X) = P + S$$
; long stock = long call + short put.
- **Greeks:** delta 0.50 → USD 1 move per USD 2 underlying; short 1,000 calls at delta 0.642 needs 642 units to hedge; gamma peaks ATM near expiry; theta −0.0250 is −2.5 cents a day; vega 0.090 and rho 0.060 are per one-point moves.
- **Eurodollar option tick = USD 25** (1,000,000 × 1/10,000 × 90/360). The 2.75% cap costing 9.25bp realises 2.6575%; the 48.25bp put realises 3.2325%.
- **Strategies:** bull call ABC 21 → long 20 @ 1.50, short 25 @ 0.75, debit 0.75, breakeven 20.75, max profit 4.25. Covered call 12.00/12.40/0.80 → breakeven 11.20, max gain 1.20, max loss −11.20.
- **Trap:** shorting a put is not a hedge — it is a bullish obligation. A collar is not riskless, and a covered call sells away the upside above the strike.

Structured Warrants and Daily Leveraged Certificates

5.1 Introduction

Structured warrants and daily leveraged certificates (DLCs) are exchange-traded SIPs. They give leveraged exposure with a smaller cash outlay, but leverage magnifies losses. Under Singapore's SIP/CAR framework, the representative must consider the client's knowledge, experience, financial situation, risk tolerance, objectives, horizon, liquidity and ability to understand the product, and document the customer-account review, education/work/investment experience.

5.2 Structured warrants

A structured warrant is issued by a third-party financial institution and references a stock, index, fund, commodity, currency or basket. It is normally shorter-dated than a company warrant; SGX warrants are generally European style unless the product terms say otherwise. A call warrant gives leveraged upside exposure; a put warrant gives leveraged downside exposure. A zero-strike warrant is economically close to a synthetic share exposure because its strike is zero.

Cash settlement per warrant is:

- call:
$$\frac{S - X}{n}$$
if positive;
- put:
$$\frac{X - S}{n}$$
if positive;

where
$$n$$
is the conversion ratio (number of warrants per one underlying unit). Physical settlement, if offered, follows the product terms.

Compared with company-issued warrants, structured warrants are issued by an unrelated financial institution and normally do not dilute the underlying company. Both can expire worthless and provide no ordinary voting/dividend rights.

Benefits include leverage, limited loss to the purchase price (subject to issuer failure), no initial margin in the normal exchange-traded purchase, transparent exchange trading, and access to multiple markets. Risks include time decay, implied-volatility changes, spread/market-maker liquidity, issuer credit, currency and corporate actions.

5.3 Key terms

- **Conversion ratio (**
$$n$$
): warrants needed for one underlying unit.
- **Strike/exercise price (**
$$X$$
): price at which the underlying is bought/sold.
- **Expiry:** final life/exercise date.
- **Issue/market price:** amount paid for the warrant.
- **Settlement:** physical or cash.
- **Gearing:** underlying price divided by
$$\text{warrant price} \times n$$
.
- **Effective gearing:**
$$\text{delta} \times \text{gearing}$$
; a better approximation of short-term percentage sensitivity than gearing alone.

The SGX code normally identifies underlying, issuer, call/put or style and maturity. Always confirm the current term sheet rather than inferring from a ticker.

5.4 Factors affecting warrant value

Life to expiry, underlying price, strike, implied volatility, dividends, interest rates, conversion ratio, liquidity, issuer credit and gearing affect price. A call rises with underlying and volatility and generally falls with dividends; a put behaves in the opposite direction for spot/dividends. Time value normally decays faster close to expiry and is not linear.

Warrant delta is
$$n \times \frac{\text{change in warrant price}}{\text{change in underlying}}$$
. Implied volatility is backed out from the observed price; it is not a guaranteed prediction.

5.5 Trading structured warrants

Calls and puts may be bought to express bullish/bearish views, hedge a portfolio or construct spreads. The investor pays the warrant price up front and normally has no daily margin call like a futures trader, but may lose the whole premium. Gearing makes the percentage movement in the warrant much larger than in the underlying. Effective gearing changes with moneyness and expiry.

5.6 Valuation

Intrinsic value:

- call:
$$\max\left[0,\ (S-X)/n\right]$$
;
- put:
$$\max\left[0,\ (X-S)/n\right]$$
.

Conversion price:

- call:
$$X + n \times WP$$
;
- put:
$$X - n \times WP$$
.

The warrant premium is its price above intrinsic value. The formula sheet also expresses dollar premium as:

- call:
$$nWP + X - S$$
;
- put:
$$nWP - X + S$$
.

Premium percentage may be calculated against the relevant conversion value; state the convention used. A call with
$$X = \$1$$
,
$$WP = \$1.20$$
,
$$n = 1$$
has conversion price \$2.20. A warrant at \$0.40 on a \$2.40 share has 6× gearing when
$$n = 1$$
.

5.7 Roles in the market

The issuer designs, funds and bears issuer risk. A designated market maker (DMM) supplies quotes and liquidity subject to product/exchange conditions, but may not quote continuously or in unlimited size in every market condition. Investors should not treat a DMM quote as a guarantee of an exit price during a suspension, extreme movement, corporate action or market disruption.

5.8 Designated market makers

The DMM normally provides bid/offer quotes and minimum quantities under the exchange's rules. Quote obligations can be suspended or adjusted for events such as underlying-market closure, price limits, trading halt, technical failure, abnormal volatility, corporate actions or insufficient information. The spread and quote size are part of the investor's liquidity risk.

5.9 Settlement

Cash-settled warrants pay the in-the-money amount according to the final settlement reference. SGX Asian-style expiry may use a five-day average or other stated observation method; the last trading day may precede expiry. Physical settlement requires delivery/payment under the contract. Do not assume the final expiry spot is one closing print.

5.10 Corporate actions

Dividends, special dividends, buybacks/cancellations, rights, bonus issues and splits can adjust strike and conversion ratio so that the product remains economically comparable. The formula sheet gives:

- dividend adjustment factor
$$\frac{P - SD - ND}{P - ND}$$
;
- buyback/cancellation factors
$$\frac{A}{B}$$
and
$$\frac{P - CD}{P}$$
;
- split factor
$$\frac{A}{B}$$
;

with the new strike = old strike × factors and new conversion ratio = old conversion ratio × factors. Read the issuer's adjustment announcement.

5.11 Exotic warrants

Exotic warrants include index warrants, currency-translated warrants, yield-enhanced/discount/capped warrants, basket warrants, commodity warrants and FX warrants. A discount/capped certificate lets the investor buy exposure at a discount but caps upside; basket/index products add correlation, composition and rebalancing risk; translated warrants add FX exposure between settlement and underlying currencies.

5.12 Daily leveraged certificates (DLCs)

DLCs are exchange-traded, short-term, high-risk products designed to deliver a daily multiple of an underlying index or asset's movement. Products include long and short certificates, commonly **3× or 5×**, with product terms allowing leverage up to **7×**; the current term sheet controls the exact multiple. They are SGD-denominated, commonly T+2, with a maximum tenor up to **three years**, and fees may include brokerage, clearing and management fees.

Daily reset/compounding means a DLC is not a simple multiple of the total return over a long period. In a volatile or mean-reverting market, repeated daily losses can erode capital even if the index ends near its starting level. It is generally unsuitable as a buy-and-hold investment.

The airbag feature may trigger when the underlying moves adversely by a stated threshold — **20% for a 3× example and 10% for a 5× example** — followed by a **30-minute suspension** and an observed reset level. The exact term sheet controls. Expiry or a zero value can terminate the investment. A DMM may quote subject to conditions.

Compared with CFDs, DLCs are exchange traded and have no OTC financing/margin account in the same form, but reset/airbag/issuer and liquidity features matter. Compared with warrants, DLCs have daily leverage targeting and usually closer-to-one directional exposure before the reset; compared with leveraged ETFs, their structure, issuer and daily reset terms differ.

5.13 Summary

Warrants embed options issued by a bank; price equals intrinsic plus time value and is shaped by conversion ratio, gearing, volatility, dividends, rates, time, liquidity and issuer credit. DLCs amplify daily returns and reset, making path dependency and compounding central risks. SIP/CAR suitability and disclosure are part of the product analysis.

Barrier Options, Binary Options and Callable Contracts

6.1 Introduction

Exotic derivatives add exercise events, path-dependent barriers, binary payoffs, baskets, Quanto conversion, unusual underlyings or callable/maturity features. Before recommending one, explain how payoff is computed, what observation triggers action, whether the option can terminate early, whether the investor can lose the premium, and who bears counterparty risk.

6.2 Barrier options

A barrier option depends on whether the underlying touches or crosses a predetermined level during a specified observation period. It is path dependent: two assets with the same expiry price may have different outcomes if one touched the barrier.

- **Knock-out/kick-out:** the option ceases when the barrier is touched; a rebate may be paid.
- **Knock-in/kick-in:** the option only becomes active after the barrier is touched.
- **Up-and-out call:** barrier above current spot; one touch can cancel the call.
- **Down-and-out put:** barrier below current spot; one touch can cancel the put.
- **Double barrier:** one upper and one lower barrier; a touch of either may knock the option out or in.
- **Rebate:** pre-agreed payment after a specified barrier event.

Barrier options are often cheaper than equivalent vanilla options because the investor gives up some payoff or accepts activation uncertainty. They are usually OTC, so counterparty, valuation, liquidity, model and documentation risks are important. If a knock-out occurs, the premium can be substantially or entirely lost.

6.3 Knock-out applications

- **Shark's-fin/barrier capital-preservation structure:** often a zero-coupon principal component plus an up-and-out call. It suits a view of a rise that will not breach the barrier. A sharp rally through the barrier may terminate the participation.
- **Barrier reverse convertible:** typically a funded note plus a short knock-in put. The coupon may be paid but capital can be converted into a falling underlying if the barrier is hit; it suits a neutral/flat view and willingness to own the underlying.
- **Bonus certificate:** a barrier below spot and bonus level above spot; if the barrier is not breached, the investor receives the bonus/underlying-linked amount, but if breached the protection/bonus may disappear. Dividends, issuer risk and downside remain.
- **Barrier straddle:** combines call and put exposure with upper/lower knock-out levels; it can express a volatility view but caps participation through barriers.

Barrier placement creates trade-offs: closer barriers are cheaper but easier to trigger; wider barriers cost more but preserve more vanilla-like participation.

6.4 Knock-in applications

Knock-in products may be cheap because the desired payoff exists only after a trigger. A knock-in put may provide protection only after a market fall; a reverse convertible's short knock-in put may create conversion risk only after a breach. The investor must understand whether the barrier is monitored continuously, at closing, intraday or on observation dates.

6.5 Application and summary of barriers

Barriers can implement capital protection, yield enhancement, a conditional hedge or a view that an asset will remain within a range. They are not "free" protection: early termination, path dependence, missed participation, rebate rules, gap risk and issuer failure can dominate the payoff.

6.6 Barrier features in structured products

Structured deposits and notes may use knock-out, knock-in, one-touch, no-touch, digital, range and callable features. Read the interaction between the deposit/bond component and the derivative: principal may be preserved only at maturity and only if the issuer pays.

6.7 Binary options

A binary/digital/all-or-nothing option pays a fixed amount if a condition is satisfied and zero (or another fixed amount) otherwise. One-touch pays if the barrier is touched; no-touch pays if it is never touched; double-no-touch/range structures pay if neither boundary is breached. Binary payoff is discontinuous and highly sensitive to barrier proximity, volatility, time and monitoring convention.

6.8 Basket options

A basket option references several assets. Correlation affects the premium: lower or imperfect correlation may make a basket option cheaper than buying each single-asset option, but it creates correlation, composition, rebalancing, dividend and concentration risk. A "worst-of" basket pays by the weakest constituent and is especially correlation-sensitive.

6.9 Knock-out warrants

Knock-out warrants combine exchange-traded warrant-style leverage with a barrier that can terminate the product before the stated expiry. A breach may cause the warrant to be worth zero or a residual amount under the terms. Investors must compare the trigger, settlement, issuer, DMM, ratio and monitoring rules with a standard warrant.

6.10 Callable bull/bear contracts (CBBCs)

CBBCs are exchange-traded leveraged long/short products, usually linked to an equity/index. A bull profits from a rise; a bear profits from a fall. Delta is often close to 1 before the call event, and gearing is:

$$\text{gearing} = \frac{\text{underlying price}}{\text{CBBC price} \times \text{conversion ratio}}$$

The **mandatory call event (MCE)** is triggered when the underlying reaches the call price: for a bear, spot rises to/above the call price; for a bull, spot falls to/below it.

- **Category N:** after an MCE there is no residual value; the product is terminated.
- **Category R:** residual value may be paid based on the issuer's post-MCE valuation window and settlement price; it may be zero.
- The call price, strike and conversion ratio are specified in the term sheet; a Category N call price need not be inferred as "always equal" to strike.

Theoretical price (simplified source formula):

- bull = $\dfrac{(\text{underlying} - \text{strike}) + \text{financial cost}}{\text{conversion ratio}}$;
- bear = $\dfrac{(\text{strike} - \text{underlying}) + \text{financial cost}}{\text{conversion ratio}}$.

Residual value for R-CBBCs:

- bull = $\dfrac{\text{settlement price} - \text{strike}}{\text{conversion ratio}}$;
- bear = $\dfrac{\text{strike} - \text{settlement price}}{\text{conversion ratio}}$.

The settlement price is subject to the MCE observation rules: a **bull** uses the **minimum** and a **bear** the **maximum** observed price between MCE and the next session, subject to the product terms. At maturity, the same direction-specific formula uses the closing settlement price.

CBBC risks include MCE/early termination, issuer credit, liquidity/DMM spread, gap risk, leverage, financing cost, underlying suspension, currency, market volatility, residual-value uncertainty and no-margin misconception. Common features with warrants include exchange trading, fixed maturity and no initial margin account; the call price/MCE is the important CBBC-specific feature.

6.11 Risks

Review path dependence, gap risk through a barrier, premium loss, model and valuation risk, counterparty/issuer risk, liquidity and secondary-market risk, correlation risk, currency risk, leverage and early termination. A barrier hit during a market gap may produce a much worse effective outcome than an investor's intended level.

6.12 Similarities and differences

Feature	Barrier option	Binary	Warrant/knock-out warrant	CBBC
Typical venue	OTC	OTC or exchange	Exchange	Exchange
Payoff	conditional vanilla/rebate	fixed/discontinuous	leveraged option	leveraged linear ex
Main trigger	barrier touch/cross	event/range/touch	product barrier, if any	mandatory call lev
Key extra risk	path/counterparty	discontinuity/monitoring	issuer/time value	early termination/

6.13 Summary

Exotics exchange a lower premium or higher coupon for conditionality. Barrier level, direction, monitoring, rebate, knock-in/out status and settlement determine the payoff. CBBCs add a mandatory call event and Category N/R distinction; investors can lose all invested capital quickly.

Chapter 6 appendix

The Chapter 6 appendix provides further barrier/CBBC applications and reinforces that a product's marketing label is not a risk description. Recalculate the payoff under spot up, spot down, barrier touched, barrier not touched, early termination, maturity and issuer-default scenarios.

Structured Deposits and Other Structured Products

7.1 Structured products: purpose and suitability

Structured products combine a deposit, note, fund, warrant, ILP, ETF/ETN or other wrapper with derivatives. They can customise a market view, enhance yield, preserve capital at a stated maturity, provide participation/discount/leverage, hedge an exposure, change currency or rebalance without directly buying the underlying.

Suitability requires understanding risk appetite, knowledge/experience, liquidity needs, investment horizon, currency, concentration, principal-loss potential, leverage, counterparty, early-exit rules, fees, tax, restrictions and the client's actual objective. Product features are trade-offs: higher coupon commonly means selling protection or accepting conversion/conditional principal.

7.2 Capital-preservation deposits

In Singapore, structured deposits are bank deposit obligations with principal paid in full at maturity, with or without interest, but they are **not necessarily covered by deposit insurance** and early withdrawal may incur loss of principal/return. The bank's credit standing still matters. "Capital preservation" normally means at maturity and subject to the stated currency and issuer — not a guarantee of mark-to-market value.

Principal component plus derivative strategies include:

- standard call/put conversion;

- knock-out deposits (feature disappears after a trigger);
- knock-in/appearing features;
- bull call/bull put spreads;
- one-touch lock-in;
- double-no-touch/range currency deposits;
- range accrual.

7.3 Yield-enhancement deposits

A standard dual-currency investment (DCI) is economically a money-market deposit plus a short currency call/put. The bank pays a higher return but can convert principal and interest into an alternate currency at the strike if the trigger condition is met. The investor must be willing to own the alternate currency and bear its value falling against the base currency.

- **KO DCI/disappearing:** the option feature terminates if a knock-out occurs.
- **KI DCI/appearing:** the conversion feature activates only after a barrier is hit.
- **Accumulator:** often long calls plus short puts, regular discounted purchases, a knock-out barrier and sometimes 1×2 gearing; downside loss can be very large.
- **Decumulator:** reverse economics; the investor may sell/accumulate the underlying through short calls and can be forced to deliver/sell when price rises.

7.4 Structured products

Range currency deposits/range accruals pay a high rate for observations inside a range and a lower rate outside it. A one-touch may lock in a rate or terminate/convert after a touch. Bull/bear spreads cap the maximum payoff. Step-up structures alter the coupon over time, often with a callable feature. Every structure should be decomposed into deposit/bond plus long or short option legs.

7.5 Summary

Structured deposits may improve yield or customise currency/market exposure but can return less than a plain deposit, convert currency at an unfavourable level, be hard to exit and expose the investor to bank credit risk. The client must understand the worst case, not only the headline yield.

Chapter 7 appendix

The non-exam appendix expands deposit strategies and demonstrates that yield enhancement is commonly the premium received for selling an option. It reinforces early withdrawal, currency conversion, barrier and suitability issues.

Structured Notes

8.1 Notes

A structured note is a debt/debenture obligation whose return is linked to an underlying equity, index, currency, commodity, rate, credit reference or basket. The investor owns a claim on the issuer (or an SPV), not the underlying asset. The note can be distributed directly by a bank or issued through an SPV; an SPV may be off-balance-sheet and may give only the legal/asset claim described in its documents.

Under Singapore offering rules, a prospectus may be required unless an exemption applies (for example accredited/institutional investors or a stated minimum investment such as **S$200,000**). Read the prospectus, PHS and term sheet.

8.2 Issuing structured notes

The principal component can be a zero-coupon bond, corporate/AAA bond or deposit-like instrument. The return component can be a long or short option, digital, range, swap, CDS or other derivative. Issuance, hedging, funding, collateral, valuation, fees, tax, maturity mismatch and issuer/SPV recourse determine the investor's actual risk.

8.3 Types

- yield with full or partial capital preservation;
- yield enhancement without capital preservation;
- capital-gain/participation notes without preservation;
- callable, inverse-floater, range, credit-linked, equity-linked, worst-of and access notes;
- ETNs and index-linked notes for exchange access.

8.4 Yield enhancement and capital preservation

Range accrual notes (RAN)

A RAN pays a high rate on observations inside a stated range and a lower rate outside. If

$$n$$

of

$$N$$

observations are inside:

$$\text{payout} = P_1 \frac{n}{N} + P_2 \frac{N-n}{N}$$

The term sheet defines reference index, range, observation frequency, payout rates, minimum/maximum, call rights, settlement and whether the condition is observed at expiry or throughout life.

Callable RANs / CDRAN / VMRAN

The issuer may call after a lockout/observation period. Calling creates reinvestment risk: the investor loses the attractive coupon and may only reinvest at lower rates. A callable deposit/return is not the same as a bond held to maturity.

Inverse floater

Coupon typically follows

$$\max\left[X\% - \text{leverage} \times \text{floating-rate index},\ \text{minimum coupon}\right]$$

. It benefits from falling rates but can fall to the floor when rates rise; leverage makes rate risk nonlinear.

Step-up callable note (CSUN-type)

Coupons step up on a schedule, but the issuer may call; investors bear issuer call/reinvestment risk, rates risk and the risk that the product's "step-up" is never received.

8.5 Yield without capital preservation

Equity-linked notes (ELNs)

An ELN is generally a funded bond/deposit plus a short put (bull ELN) or short call (bear ELN), and can include a strangle/worst-of structure. A bull ELN pays a higher yield if the equity stays above the strike; if below, the investor may receive shares/cash at the strike and bear equity downside. The principal can fall to zero if the underlying collapses. Cash versus physical settlement, knock-in/out, conversion ratio, observation and issuer terms matter.

Worst-of ELNs are driven by the weakest underlying and are highly sensitive to correlation, volatility and the weakest name's liquidity.

8.6 Credit-linked notes (CLNs)

An investor in a CLN effectively sells credit protection through a CDS on a reference entity or basket and receives an enhanced coupon. On a credit event—bankruptcy, failure to pay, restructuring, obligation default/acceleration or repudiation/moratorium as defined—the payout may be reduced, securities delivered or principal lost. The investor bears both issuer/SPV credit risk and reference-entity credit risk. First-to-default baskets amplify correlation/concentration risk.

A bond-linked note (BLN) embeds a short put on a bond and has issuer/bond price risk; it is not identical to a CLN's CDS credit-event mechanics.

8.7 Market access

- **ETN:** unsecured listed debt tracking an index; no ordinary ownership and no coupon necessarily; issuer credit risk is central.
- **ETF:** pooled fund with assets/custody and creation/redemption; it can still have market, tracking, liquidity and counterparty risks.
- **Index-linked note:** participation/minimum/cap linked to an index, normally subject to issuer claim.
- **Participation/access note (PN):** gives exposure to a market or asset through a note rather than direct ownership; terms and issuer recourse control.

8.8 Investors and suitability

Assess liquidity and lock-up, risk/return, capital protection conditions, underlying scenario, term, fees/charges, currency, issuer/SPV, collateral, secondary market, early termination, taxes and concentration. Explain that a high coupon can be compensation for selling a put/call/credit protection.

8.9 Summary

Structured notes are debt plus derivatives. Ask whether principal is actually protected, when, in which currency and by whom. Callable, range, inverse-floater, ELN and CLN products shift market, rate, equity or credit risk into the investor's return; issuer/SPV and liquidity are as important as the reference asset.

Chapter 8 appendix

Further coverage includes spread CDRANs, RAN variations, documentation and PHS content. A PHS should make the product understandable through a product summary, suitability, key features, key risks, prominent/clear disclosure of possible loss of all principal, definitions, contact information, readable layout and simple language. Appendix B reproduces a MAS practice-note excerpt: the PHS is prepared under the SFA framework, submitted with the prospectus at lodgement, made available with/alongside the prospectus (including online), and follows the prescribed format/headings as a minimum while adding material information needed for investor understanding.

Structured Funds

9.1 Structured funds

A structured fund combines securities and/or derivatives to create a specified risk-return profile, capital-preservation objective, participation formula, yield or cost structure. Capital preservation normally applies only at maturity and is subject to fund/issuer/strategy performance; it is not the same as deposit insurance or a guaranteed NAV. Funds are governed by the applicable Code on Collective Investment Schemes, offering documents and licensing/custody requirements.

Key fund governance points are a licensed manager, fiduciary trustee/custodian, NAV calculation/reporting, valuation and pricing policies, disclosure of fees/risks, audit and investor dealing terms. Conflicts may arise where the manager, issuer, swap counterparty, custodian or distributor is affiliated.

Core fund vocabulary

Know the glossary terms: asset allocation, AUM, closed/open-ended fund, CIS, CPPI, cushion, custodian, drawdown, excess return, index fund, ISIN, management fee, minimum subscription, multiplier, NAV, option,

share class, Sharpe ratio, SICAV, TER, total return, total return swap (TRS), tracker, trailer/upfront sales charge, UCITS, umbrella fund, underlying, VaR, volatility and zero-coupon bond.

Traditional, tracker and structured funds

Traditional active funds seek to select assets; trackers seek to replicate an index; structured funds use formulae/derivatives to transform the distribution. Evaluate the underlying, maturity, payout, market outlook, fees, liquidity, counterparty and tax—not only the fund label.

Total return swap (TRS) and option structures

A TRS fund receives the total return of a reference asset/index and pays financing to a swap counterparty. It provides efficient exposure but adds counterparty, collateral, reset, liquidity and termination risk. A "zero-plus" structure typically combines a fixed-income/zero-coupon component with a call, offering a floor-like maturity outcome and upside participation subject to costs and issuer/fund risk.

CPPI

Constant proportion portfolio insurance allocates between risky and risk-free assets:

$$\text{cushion} = \text{portfolio value} - \text{floor value}$$

$$\text{multiplier} = \frac{1}{\text{crash size}}$$

$$\text{risky allocation} = \text{multiplier} \times \text{cushion}$$

$$\text{risk-free allocation} = \text{portfolio value} - \text{risky allocation}$$

As the portfolio rises, risky allocation normally rises; as it falls, the manager sells risky assets and moves into the risk-free asset. The floor may increase over time and the risky allocation may be capped at 100% or allowed to be leveraged depending on the terms. A sharp fall can force the fund entirely into the risk-free asset, after which it may miss a later recovery: this is the "cash-lock" or gap-to-floor risk. In a range-bound market CPPI may buy high and sell low. DPPI varies the multiplier and has similar path, fee, gap, liquidity and manager risks.

Other structured-fund arrangements

Common forms are capital-preservation funds, formula funds, capitalised/distribution classes and direct/indirect swap-based structures. Commodity-linked funds must consider futures roll yield: backwardation can help roll return; contango can hurt it. Option-based funds must explain the premium, strike, cap, barrier and issuer/counterparty.

Roles, controls and after-sales

The manager implements the mandate; trustee/custodian safeguards assets and oversight; administrators calculate NAV; auditors review; distributors provide information/suitability. Controls include operational, financial, legal, independent trustee, valuation, liquidity and conflict controls. Documents include prospectus, PHS and trust deed. Investor service includes dealing/NAV, factsheets, monthly reports, tax and corporate-action information.

Investors may be retail, accredited or institutional. Core-satellite allocation, rebalancing, liquidity management and cash investment are common uses. Assess tracking error, leverage, derivative counterparty, concentration and redemption gates.

9.2 Structured ETFs

An ETF is typically an open-ended listed fund that trades intraday and uses authorised participants to create/redeem units. Types include equity, fixed-income, cash, currency, commodity, index, alternative, leveraged, inverse and protective ETFs.

- **Physical/direct replication:** holds all or a representative sample of securities.
- **Synthetic replication:** uses swaps/derivatives; a **10% net counterparty-exposure** concept applies subject to the rules and documentation — collateral is the key control.

Pricing risks include bid–ask spread, NAV discount/premium, tracking error, TER, roll yield, cash drag, FX, market liquidity and counterparty collateral. Creation/redemption arbitrage helps keep price near NAV but can fail in stressed markets. ETFs provide intraday liquidity, diversification, transparency, short/leveraged/inverse exposure and core-satellite tools, but are not risk-free. Compare ETFs with ETNs (unsecured debt), mutual funds (usually NAV dealing) and listed derivatives.

Investors should understand index methodology, rebalancing, distributions, securities lending, swap/collateral, leverage reset and closure/merger risk. After-sales monitoring includes NAV, tracking difference, corporate actions and factsheets.

9.3 Summary

Structured funds are CIS vehicles whose payoff depends on assets, formulae and sometimes derivatives or swaps. CPPI preserves a floor only subject to path and issuer/fund risks. ETFs trade intraday but have NAV, tracking, market, liquidity, FX and counterparty exposures. Governance, custody, disclosure and suitability are part of the product.

Contracts for Differences (CFDs)

10.1 History

CFDs emerged in the UK in the 1990s from equity-swap-style products and reached retail markets in the late 1990s; they were **not** available in the US retail market in the same way. That history explains their OTC, leveraged and cash-settled character.

10.2 Definition

A CFD is an agreement to exchange in cash the difference between the opening and closing value of an underlying position (or the reverse for a loss). The investor does not own the share/asset, normally has no voting rights, and does not take physical delivery. Long equity CFDs may receive a dividend adjustment; short positions may be debited. Corporate-action adjustments follow the provider's terms.

10.3 Characteristics

Leverage, flexibility, relatively low initial outlay, transparent reference price, simplicity of long/short exposure, no fixed expiry in some provider models, global access, cash settlement and corporate-action adjustments are key features. Countervailing risks are margin calls, forced liquidation, financing, commission/GST, rollover, spread, provider credit and market liquidity.

CFD terms include underlying, opening/closing price, number of units, direction, currency, margin, financing rate, expiry/rollover and corporate-action treatment.

10.4 Business models

- **Market maker:** provider quotes the CFD and may internalise/hedge flow; price and spread depend on its ability and discretion.
- **Direct market access (DMA):** orders route through the provider to the underlying market; DMA is **predominant in Singapore**.
- **Exchange-traded CFD:** exchange/clearing structure can reduce bilateral counterparty risk but remains product-specific.

10.5 Advantages

CFDs allow long or short access to shares, indices, sectors, FX, commodities, government treasuries, ETFs, VIX/volatility references, real estate and other global instruments from one account, with a smaller initial cash outlay and no need to own/borrow the underlying in the same way as a cash trade. They can be used for hedging, tactical exposure, pairs trading and dividend capture.

10.6 Mechanisms

10.6.1 Opening and margin

Initial margin is a percentage of contract value. The investor controls a larger notional than the cash posted; return on margin can be high but losses are based on the full exposure. Maintenance margin and provider limits apply. The Risk Disclosure Statement (RDS) should explain margin, liquidation and the client's liability.

10.6.2 Mark-to-market and margin calls

Positions are revalued as the underlying moves. A margin call requires additional funds; if not received, the provider may sell some or all positions. The investor remains liable for a shortfall, loss and expenses after liquidation. A stop-loss is not a guarantee against gap/slippage.

10.6.3 Financing and corporate actions

Long CFDs normally incur interest on the full notional; short CFDs may receive interest on deferred sale proceeds but can incur stock-borrow costs. Commissions are commonly a percentage of notional subject to a minimum; GST may apply. Rollover fees depend on provider and asset. Long equity positions may receive cash dividends; short positions pay them. Splits/reverse splits adjust quantity/price; scrip dividends, bonus and rights entitlements may differ or require close-out.

10.6.4 Orders and stops

All orders are day orders unless otherwise stated and use the underlying bid/ask. Limit orders transact at the stated price or better. MTL orders execute at the market and leave an unfilled balance at the prior execution price. Market orders execute at the best available bid/ask. Stop-loss and stop-entry orders activate at a trigger; fast markets can gap through the stop. A guaranteed stop, if offered, is a distinct paid service.

10.6.5 Cost examples

Worked example. Buy 10,000 shares at \$2, sell after 10 days at \$2.20, with 0.4% commission, 8% GST on commission and 5% annual financing. Buy commission/GST = \$86.40; sell = \$95.04; ten-day financing = \$27.78; total expense = 209.22; $netprofit = **1{,}790.78$**.

Short example: sell 10,000 at \$3.30, cover at \$3.10, 0.4% commission, 8% GST, 20% margin and 6% financing for five days. Total expense = \$303.98; net profit = \$1,696.02; capital posted = \$6,600; ROI 25.70%. If the price rises to \$3.45, expense = \$319.10; net loss = \$1,819.10; ROI −27.56%.

10.6.6 CFD versus futures

CFDs are mostly OTC, can be extended/rolled subject to provider policy, explicitly charge financing, may pass dividend adjustments and carry provider counterparty risk. Equity futures are exchange-traded, have fixed expiry, embed financing in the price, normally have no dividend entitlement and are also margined/leverage products.

10.7 Global trading opportunities

Underlyings include Singapore **STI/SiMSCI** and global equity/sector indices (S&P 500, DJIA, FTSE100, CAC40, DAX, Nikkei225, ASX200, Hang Seng, China A50), currencies (AUD, CAD, EUR, CNH, JPY, CHF, GBP, USD), energy/industrial/precious metals/agricultural commodities, US/UK/Japan/Euro government bonds, ETFs such as SPDR Gold and volatility/real-estate references. Availability is provider-specific.

10.8 CFD trading strategies

- **Dividend capture/stripping:** buy before ex-dividend and close after, seeking to benefit if the price does not fall by the full dividend; costs, tax, timing and price behaviour can defeat the trade.
- **Pairs trading:** long an undervalued/stronger asset and short an overvalued/weaker related asset. It aims for mean reversion and market neutrality but correlation can break and costs double.
- **Industry-sector pairs:** compare companies such as property stocks; stock-specific and sector risks remain.
- **Statistical arbitrage:** use historical correlation/quantitative signals; a relationship may not revert.
- **Merger arbitrage:** long target/short acquirer or the reverse; deal failure can hurt both legs.
- **Index pairs:** trade correlated indices such as S&P500 and Nikkei225 when their relationship diverges; index liquidity is usually deeper than single stocks.

10.9 Summary

CFDs are leveraged cash-settled contracts, not ownership. Their flexibility hides financing, margin, liquidation, counterparty, liquidity, currency, corporate-action and short-sale risks. Learn the provider model, order mechanics, total cost and worst-case liability.

Key Products and Investment Risks

11.1 Investment risks for derivatives

Futures

- **Basis risk:** cash and futures are not identical or do not move together; causes include asset mismatch, uncertain exposure date, early close-out and contract-size mismatch/under- or over-hedging.
- **Liquidity risk:** volume/open interest may be low; price limits/lock-limit markets or near-expiry conditions may prevent offsetting.
- **Leverage/margin risk:** loss can exceed initial margin; a call may force liquidation and a shortfall remains payable.

Options

Holders can lose the entire premium, especially when OTM and near expiry; European options may be difficult to monetise before expiry, and OTC exercise provisions may be unusual. Naked writers can face unlimited/large losses, and American writers do not control exercise timing. Both sides face volatility, complex-strategy, trading-halt, foreign-market timing and currency risks.

Equity-index options may reference illiquid/niche indices and OTC options add counterparty risk. Bond options inherit bond rates, credit-spread, liquidity, embedded-option and sovereign risks. Currency options involve huge notionals, round-the-clock market movements, currency delivery and OTC counterparty risk.

Warrants

Company warrants can expire worthless, give no voting/dividend rights and dilute shareholders if exercised. Structured warrants add issuer credit risk, underlying market risk, leverage, time decay, volatility, liquidity and currency exposure.

CFDs

Leverage, market volatility, counterparty, liquidity, currency, financing, commission/GST, rollover, corporate actions and provider-model risk apply. Foreign markets may have weaker protections and exchange controls. Market-maker pricing may be less transparent than DMA.

11.2 Key risks of structured products

Structured products combine risks that are not obvious from a headline yield:

1. **Market risk:** macro rates, inflation, employment, recession, earnings and political change affect underlying/payoff. Selling a call sells protection against a rise; selling a put sells protection against a fall; a short receiver/payer swaption sells protection against a rate move.

2. **CPPI risk:** buys more risky assets as they rise and sells as they fall; a sharp fall can lock the fund at its floor.
3. **Credit/counterparty risk:** issuer, SPV, swap counterparty, reference obligor or collateral may fail. An SPV may not be guaranteed by its parent.
4. **Credit-event risk:** bankruptcy, obligation acceleration/default, failure to pay, repudiation/moratorium and restructuring can trigger CDS/CLN losses.
5. **Liquidity/secondary-market risk:** customised options/CDS may be illiquid; lock-up and maturity mismatch may force a discount sale.
6. **Early-termination risk:** unwinding the bond and derivative before maturity incurs discount and break costs; full principal preservation may exist only at maturity.
7. **Structure risk:** combinations vary; loss may be unlimited or principal may be fully exposed despite a familiar wrapper.
8. **Reinvestment risk:** coupons/principal returned early or through calls may be reinvested at lower rates; large coupons, frequent payments, long maturities and premium bonds increase it; zero-coupon bonds reduce interim reinvestment risk.
9. **Volatility risk:** embedded options change value with volatility; a callable bond can fall when the call option becomes more valuable.
10. **Correlation risk:** basket/worst-of/credit-pool constituents may fall/default together.
11. **FX risk:** underlying and note currencies may differ.
12. **Legal risk:** investors own the note/claim, not necessarily underlying assets; creditor priority/recourse depends on documents.
13. **Transactional risk:** pricing/contract commencement and settlement timing, especially across currencies, can create exposure.
14. **Mis-selling/incongruence:** the product may not meet the client's objective; PHS/prospectus/private-placement documents and advice must explain the real payoff.

11.3 Trading risk control and management

The front office, middle office and back office manage market, credit and settlement risks.

Futures controls

- **Open-contract limit:** smaller for volatile/illiquid products and restricted relative to gross open interest; separate outright/spread limits may apply.
- **Maximum-loss limit:** weekly/monthly trader or desk loss triggers squaring-off and suspension/escalation.
- **Maturity limit:** restricts illiquid deferred contracts.
- **Stress-test limit:** measures portfolio under adverse scenarios.

For fixed income: 1 bp = 0.01%; DV01/PVBP is dollar price change per 1 bp; duration is a weighted cash-flow time measure; modified duration measures price sensitivity:

$$\frac{\Delta P}{P} \approx -\text{modified duration} \times \Delta YTM$$

. A modified duration of 2 and a 100 bp yield rise implies about a 2% price fall. If price $980 has PVBP $1, a 5 bp yield rise reduces it to $975.

VaR combines volatility/correlation into a loss threshold. A 1-day 95% VaR of SGD1m means a 5% probability that loss exceeds SGD1m over one day, under the model's assumptions. VaR has confidence, horizon and loss-estimate components; it is not a guarantee or stress-test substitute.

Options controls

Set limits for delta (currency exposure), gamma (delta change for a spot move), vega (premium change per 1% volatility), theta (time decay) and rho (rate sensitivity). Gamma is especially volatile near ATM and expiry. Maturity concentration, total book size, premium payment and intraday limits complement Greek limits. Gamma, vega and theta may be managed together for short-option books.

Credit and settlement controls

OTC counterparty limits depend on credit rating, settlement amount and tenor. A Credit Support Annex (CSA) specifies collateral. Clearing-broker risk arises because margin is held with the broker; exchange/CCP

risk remains despite default resources. Back-office settlement controls require timely, accurate confirmations including option style, expiry/date/time, premium payment date, settlement date, strike, currencies and amounts.

11.4 Summary

Risk management must match the product: basis/liquidity/leverage for futures; premium/exercise/volatility/Greek risk for options; issuer/gearing for warrants; margin/financing/provider risk for CFDs; and layered market, credit, liquidity, legal, structure, correlation, FX and reinvestment risk for structured products.

Case Studies

These cases are decision exercises, not product endorsements. A representative should establish suitability, knowledge, risk willingness, financial capacity, external exposure, internal limits, buffers, leverage, stop/offset discipline, profit targets and current market information before recommending a trade.

12.1 Enhanced yield with currency investments — Mrs Goh

Mrs Goh holds USD, GBP and SGD. She expects a short-term USD rebound but longer-term USD weakness and accepts conversion into SGD/GBP. A USD/SGD DCI invests USD1m for three months at 11% p.a., strike 1.3800, spot 1.3569.

- If expiry spot is 1.3300, no conversion at strike; USD proceeds = 1,000,000 + 1,000,000 × 11% × 3/12 = USD1,027,500. Converted at 1.33 = SGD1,366,575.
- If spot is **1.3800**, proceeds remain USD1,027,500.
- If spot is **1.4100** and conversion applies, USD1,027,500 is converted at 1.3800 = SGD1,417,950.

From the bank's perspective, fund the deposit at 1.3919% and sell the option at 0.0250, giving about **USD1,028,566.74** after three months — annualised around **11.43–11.44%**, above the 11% client return. DCI yield is option premium plus deposit interest and conversion risk.

The GBP/USD DCI example at 10% p.a., strike 1.3000 shows that if spot is 1.2800, the conversion calculation yields about USD1,009,230.76; if spot is 1.2600, proceeds about USD993,461.54 imply annualised return around −2.62%. A high headline yield does not prevent capital loss in the base currency.

12.2 Short-term downside volatility — Brian

Brian expects a two-month equity correction, higher volatility and longer-term recovery. Before trading, verify risk willingness/understanding, capacity and external positions, experience, internal limits, buffers, leverage, stop/offset plans, profit taking and market information. Limits, margin buffers and education help; maximising leverage does not.

Possible tools:

- long index/futures puts: limited premium loss, direct downside insurance, time decay and expiry risk;
- put warrants: similar limited-premium downside exposure, time decay/issuer risk;
- short index futures: direct daily-settled exposure, margin and potentially large losses if the market rises;
- short or long inverse/leveraged DLCs: daily reset and amplified risk;
- inverse ETFs/ETNs: exchange access but tracking, cost, roll and leverage risk;
- long volatility futures ETF: benefits from a volatility spike but suffers from leverage and contango/roll decay; long-term holding can trend toward zero.

Timing is difficult; short products are costly to maintain and should be closed/rolled with discipline.

12.3 Portfolio protection — William

William holds OCBC, DBS and CapitaLand worth SGD92.72m and wants protection beyond a 7% fall. Compare put-warrant strikes, breakeven, expiry and stock coverage. In the 7% scenario, DBS/OCBC put warrants B and D are in-the-money; CapitaLand E is selected because it is the only available CapitaLand protection and is less OTM than alternatives. A and D offer more time protection; E covers CapitaLand.

Higher volatility raises warrant prices because the strike is more likely to be reached. OTC alternatives are lending the shares for a 4–5% fee, buying tailored protective puts 3–4% below spot, or shorting MSCI Singapore futures. The futures hedge is imperfect because the three stocks represent only about 31% of the index. Futures have immediate daily MTM obligations; a long put has a right, not an obligation; a share loan earns a fee.

12.4 ELN yield and discounted blue-chip purchase — Mr Sunjoyo

OCBC spot = SGD13.30; ELN purchase = SGD975,000; notional = SGD1m; strike = SGD12.50; 80,000 shares; three months; annualised headline yield = 10.66%.

- At expiry SGD12.40, shares are worth SGD992,000; return over three months = 1.7435%; annualised compound return 7.16%.
- At SGD12.00, shares are worth SGD960,000; return = −1.5385% over three months; annualised −6.29%.
- Breakeven = 975,000/80,000 = SGD12.1875 SGD12.19.

The client is effectively short a put: below the strike, he owns the falling stock and can lose up to almost the entire investment if the stock goes to zero.

12.5 Range accrual note — Robert

AUD/USD range = 0.7600–0.8200, maturity three months, maximum yield 6%, minimum 1%, MMR 2.75%. If inside for two months and outside for one, annualised payout is

$$6\% \times \tfrac{2}{3} + 1\% \times \tfrac{1}{3} = 4.33\%$$

(the answer text displays approximately 4.32% due to rounding).

To make the digital spread cheaper or raise maximum yield: narrow the range, skew it so spot is nearer a barrier, extend maturity, or lower the minimum yield. Widening the range makes inside outcomes more likely and the option more expensive; shortening maturity can increase cost. A double-no-touch version observes the path: touching either barrier terminates the enhanced payout and leaves only the minimum return.

12.6 Equity accumulator — Harry

ABC spot SGD1.10; strike SGD1.00; knock-out SGD1.30; one year/250 business days; 10,000 shares per observation; monthly settlement. An accumulator combines long call and short put economics and can be funded or margin/unfunded. Below strike under a 1×2 arrangement, the client may have to buy 2×; above strike but below the knock-out, 1×; at/above the barrier, the note terminates.

- If the barrier is touched on day 13, days 1–12 accumulate: 120,000 shares.
- Twenty days in month 1 plus 12 days in month 2 before day-13 knock-out = 320,000 shares; cost = SGD320,000.
- If marked at SGD1.30, value = SGD416,000 and gain = SGD96,000.

Worst case: price zero for 250 days, 2.5m shares purchased at SGD1 = **SGD2.5m loss**. There is no capital preservation; the bank may margin-call/liquidate; gains are capped by the knock-out while losses continue below strike; early-break costs can be large.

12.7 Gold, silver and USD volatility — Brandon

Rising US rates may strengthen USD and weaken gold/silver, while political/equity uncertainty may support safe-haven gold. Gold's widening premium to silver and different USD sensitivities create relative-value ideas. If the direction is uncertain but a large move is expected, buy a gold straddle: long call and long put at the same strike. A strangle uses different OTM strikes and is cheaper but needs a larger move. Combine a bull call spread with a bear put spread, or a bull call with a bear call, to express two-sided volatility; do not use two bearish or two bullish structures if the view is genuinely two-sided.

12.8 Yield-curve movement — Tracy

Tracy expects the US yield curve to flatten/invert: short rates rise faster than long rates. A constant maturity swap (CMS) resets a floating leg against a long-term constant-maturity rate; unlike a vanilla IRS,

the reference maturity/duration remains constant. It can express steepening/flattening views. To benefit from short rates rising faster, swap the long end for the short end — pay the long-end rate and receive the short-end rate — subject to the exact swap convention and market quotation.

Exam cues

The case studies test the same method, not the products. For each, work the numbers and identify the embedded option and the investor's obligation.

- **DCI (Goh):** yield = option premium + deposit interest; conversion risk is the real exposure. A high headline yield does not prevent a base-currency loss.
- **Downside volatility (Brian):** puts / put warrants cap loss at premium but decay; short futures and inverse/leveraged products add margin, daily reset and compounding risk.
- **Portfolio protection (William):** compare put-warrant strikes, breakeven and coverage; a futures hedge is imperfect when the basket is only part of the index; a long put is a right, a futures position is an obligation.
- **ELN (Sunjoyo):** effectively short a put — breakeven = purchase / shares; below strike the investor owns the falling stock and can lose almost everything.
- **Range accrual (Robert):** payout = max × (days inside / total) + min × (days outside / total); a double-no-touch observes the path and a barrier touch leaves only the minimum.
- **Accumulator (Harry):** 1× above strike to the knock-out, 2× below strike; gains capped by the knock-out while losses continue — worst case is multiples of the notional.
- **Two-sided volatility (Brandon):** a straddle/strangle for a large move; combine a bull call spread with a bear put spread — never two structures on the same side.
- **CMS (Tracy):** pay the long end and receive the short end to profit from short rates rising faster than long rates.

Appendix A — CPPI strategy worked examples

Parameters: initial $100; crash size 20%; multiplier

$$1/20\% = 5$$

; risky asset S&P index fund; five-year tenor.

- Day 1 floor 85%: cushion 15; risky = 5×15 = $75; risk-free = $25.
- New floor 86% with total still $100: cushion 14; risky = $70; risk-free = $30.
- Portfolio at $86 after risky asset falls 20%: risky falls $70→$56; plus $30 risk-free = $86; liquidate all risky into risk-free, so later equity appreciation is missed.
- Portfolio at $96: risky value $66, required risky allocation

  $$5 \times (96 - 86)\% = \$50$$

  ; sell $16 risky and hold $46 risk-free.
- Portfolio at $110: required allocation

  $$5 \times (110 - 86)\% = \$120$$

  ; if capped, put all $110 in risky; if leverage permitted, borrow $10 and hold $120 risky.
- If leveraged risky value falls 20% from $120 to $96, repay $10 and move remaining $86 to risk-free. The crash has again locked the floor.

The reallocation repeats periodically. CPPI assumes an appreciating, relatively low-volatility asset with limited drawdown; a range-bound or gap-down market can cause buy-high/sell-low behaviour.

Appendix B — MAS Product Highlights Sheet practice note

An excerpt of **MAS Practice Note SFA PN-02** (issued 8 June 2018), referencing SFA 13-G13. Key points:

- the note guides issuers and professional advisers in preparing a PHS under the SFA framework and is read with the relevant offers regulations;
- the PHS is submitted to MAS at prospectus lodgement;

- it is made available with the prospectus, including online if the prospectus is online;
- the prescribed format, tabular structure, headings/subheadings and yellow Pantone 109C right-edge feature are minimum presentation requirements;
- additional headings and material information should be added where needed so investors understand the product;
- a PHS should plainly state product summary, suitability, key features, risks, fees/charges, definitions and contacts, and prominently disclose possible loss of all principal where applicable.

Appendix C — Futures contract specifications

These specifications were updated as at **April 2022**; always use current SGX specifications.

Contract	Main terms to remember
SGX EUR/CNH FX futures (`EC`)	EUR100,000; tick CNH10; minimum fluctuation CNH0.0001; cash-settled in CNH; last
SGX 10-year full-size JGB (`JG`)	JPY100m notional, 10-year 6% coupon; quarterly March/June/September/December
SGX Nikkei 225 (`NK`)	JPY500 × index; 6 serial + 32 quarterly months; outright tick 5 points/JPY2,500, str
SGX FTSE Taiwan (`TWN`)	USD40 × index; 2 serial + 12 quarterly months; 0.25-point/US$10 standard tick; 10%
SGX MSCI Singapore (`SGP`)	SGD100 × index; 2 serial + 4 quarterly months; 0.05 point/SGD5 tick; ±10%/±15%
SGX STI (`ST`)	SGD10 × index; 2 serial + 4 quarterly months; 1 point/SGD10 tick; +10%/+15% coo

Exam habit: identify contract size, tick value, trading hours/session, price limits, last trading day, settlement basis, final settlement reference and position-accountability threshold before calculating.

Appendix D — Formulae sheet (consolidated)

Futures

$$F_{0,t} = S_0(1 + C_{0,t})$$

$$P = 100(1 - R)$$

for short-rate futures.

$$\left(1 + \frac{d_1 r_1}{360}\right)\left(1 + \frac{d_2 r_2}{360}\right) = \left(1 + \frac{DR}{360}\right)$$

for long-rate futures.

$$F = S \times \frac{1 + R_c(n/360)}{1 + R_b(n/360)}$$

for interest-rate parity.

$$F = S\,[1 + r(t/360)] - D$$

for equity-index futures.

$$\text{target hedge rate} = \text{futures rate} + \text{target-rate basis}$$

$$V_t = S_t + (F - F_t) = F + (S_t - F_t)$$

$$\text{bond hedge ratio} = \frac{\text{PVBP(hedged security)}}{\text{PVBP(CTD)}} \times \text{conversion factor}$$

$$h = \frac{\Delta S}{\Delta F}$$

; contracts for a delta-neutral loan

$$= \frac{-h \times \text{loan value}}{\text{contract size}}$$

$$N = \frac{VP}{F \times T} \times \beta$$

for equity futures hedging.

Options

$$\text{option price} = \text{intrinsic value} + \text{time value}$$

$$\text{call IV} = \max(0, S - X)$$

;

$$\text{put IV} = \max(0, X - S)$$

$$C + \text{PV}(X) = P + S$$

$$\text{delta} = \frac{\text{change in option price}}{\text{change in underlying}}$$

;

$$\text{gamma} = \frac{\text{change in delta}}{\text{change in underlying}}$$

Warrants and CBBCs

$$\text{gearing} = \frac{\text{share price}}{\text{warrant price} \times n}$$

;

$$\text{effective gearing} = \text{delta} \times \text{gearing}$$

$$\text{call IV} = \max[0, (S - X)/n]$$

;

$$\text{put IV} = \max[0, (X - S)/n]$$

$$\text{call conversion price} = X + nWP$$

;

$$\text{put conversion price} = X - nWP$$

$$\text{call cash settlement} = (S - X)/n$$

;

$$\text{put cash settlement} = (X - S)/n$$

$$\text{bull CBBC} = \frac{(\text{underlying} - \text{strike}) + \text{financial cost}}{\text{conversion ratio}}$$

$$\text{bear CBBC} = \frac{(\text{strike} - \text{underlying}) + \text{financial cost}}{\text{conversion ratio}}$$

R-CBBC residual: bull

$$(\text{settlement} - \text{strike})/n$$

; bear

$$(\text{strike} - \text{settlement})/n$$

; N-CBBC residual is zero at MCE.

Structured products

$$\text{CPPI cushion} = \text{portfolio} - \text{floor}$$

;

$$\text{multiplier} = 1/\text{crash size}$$

;

$$\text{risky} = M \times \text{cushion}$$

;

$$\text{risk-free} = \text{portfolio} - \text{risky}$$

$$\text{RAN payout} = P_1 \frac{n}{N} + P_2 \frac{N-n}{N}$$

$$\text{inverse floater coupon} = \max\left[X\% - \text{leverage} \times \text{floating index},\ \text{minimum coupon}\right]$$

$$\text{participation rate} = \frac{\text{discount sum}}{\text{call premium}}$$

under the structured-product convention.

Appendix E — Review Questions: what they test

Appendix E is for familiarisation with exam scope and format, not a complete study resource. Its 50 questions cover futures (1–5), futures strategies (6–10), options (11–15), structured warrants/DLCs (16–21), barriers/binaries/CBBCs (22–26), structured deposits/products (27–31), structured notes (32–36), structured funds (37–40), CFDs (41–45) and risks (46–50). The expanded bank in `CMSIP_QUESTIONS.md` follows this order, adds Chapters 1 and 12, and covers every subsection in this guide.

Appendix F — Essential readings

Recommended: Bodie, Kane & Marcus, *Investments*; Hicks, *Managing Currency Risk Using Foreign Exchange Options*; Walmsley, *Foreign Exchange and Money Markets Guide*; Fabozzi & Leibowitz, *Fixed Income Analysis*; the CME Group *Risk Management Handbook*; Chance, *Analysis of Derivatives*; Hull, *Options, Futures & Other Derivatives*; Kolb & Overdahl, *Financial Derivatives: Pricing and Risk Management* and *Futures, Options, and Swaps*; Wilmott, *Derivatives*; Blumke, *How To Invest in Structured Products*; Lancaster, Schultz & Fabozzi, *Structured Products and Related Credit Derivatives*; Swiss Structured Products Association; Fabozzi, Davis & Choudhry, *Introduction to Structured Finance*; and current MAS, MoneySense and SGX materials.

Appendix G — Glossary and cross-reference

General finance language used throughout the chapters. The most important terms are defined in context above; use this compact cross-reference when revising:

- **Market basics:** absolute/relative return, alpha, benchmark, beta, bid/ask, blue chip, asset class, asset allocation, diversification, position, portfolio, risk appetite, risk tolerance, systematic/specific risk, spot market/rate, primary/secondary market, supply/demand and price discovery.
- **Accounting/economics:** accounts payable/receivable, balance sheet, cash-flow statement, current assets/liabilities, equity, book value, net income/profit margin, ROA/ROE, EBITDA, GDP, inflation/deflation, recession, fiscal/monetary policy, business cycle, balance of payments, current/capital account, exports/imports, trade balance, goodwill, amortisation, depreciation, fixed/variable costs, WACC, DCF/NPV, EPS, P/E/P/B/P/S/P/CF, enterprise value and EV/S.
- **Derivatives:** derivative, forward, futures, options, premium, strike/exercise, call, put, American/European, ITM/ATM/OTM/moneyness, intrinsic/time value, delta, gamma, theta, vega, rho, compound/exotic/plain-vanilla option, options on futures, warrant, structured warrant, conversion ratio, gearing, effective gearing, CBBC, DLC, CFD, swap, swaption, IRS, currency/commodity/equity

swap, CDS, cap, floor, collar, basis, basis point, carry, mark-to-market, initial/maintenance/additional margin, notional, delivery, settlement, contract specification, hedge/hedge ratio, stack/strip, spread, calendar/horizontal, vertical, diagonal, butterfly, condor, ratio and seagull.

- **Structured products/funds:** structured product, structured deposit, structured note, ELN, CLN, BLN, RAN, CDRAN/VMRAN, inverse floater, zero-plus, accumulator/decumulator, discount/capped certificate, participation rate, basket/worst-of, barrier, knock-in/knock-out, one-touch/no-touch, CPPI, cushion/multiplier, DPPI, TRS, zero-coupon bond, ETN, ETF, NAV, AUM, TER/expense ratio, Sharpe/Treynor/information ratio, tracking error, UCITS, CIS, SICAV, umbrella, open/closed-ended fund, custodian, trustee, prospectus, PHS, trust deed, accredited/institutional/retail investor.
- **Risk/control:** market, basis, leverage, liquidity, credit/counterparty, settlement, operational, legal, transaction, structure, correlation, volatility, currency, financing, reinvestment, early termination, mis-selling, sovereign, political, model, gap and roll risk; VaR, stress test, open-contract/maturity/maximum-loss/Greek/book/intraday limits, CSA, front/middle/back office, DMM and authorised participant.
- **Corporate/market terms:** dividend/stock dividend, bonus/rights issue, share split/reverse split/buyback, conversion value, straight value, convertible bond, current yield/YTM, credit spread/rating, default/credit event, repo, T-bill/T-bond, MBS/ABS/CDO, dark pool, ECN, dealer/market maker, brokerage, proprietary trading, hedge fund, private equity, venture/growth/buyout/distressed funds, REIT, real estate, alternative investment, index/price-weighted/capitalisation-weighted, index fund, passive manager, technical analysis, EMH, APT, CAPM, efficient frontier, GIPS and information ratio.

The less product-specific entries — annuity, appraisal, appreciation/depreciation, assets, banks, board, foundations, fraud, insurance, legal/regulatory terms, policies/procedures/processes, reserve currency/requirement, retained earnings, shareholders/voting rights, tariffs, underwriters, trade bills, trust deed and similar general terms — are ordinary definitions supporting the product chapters. When a question uses one, translate it back to the underlying cash flow, risk or legal relationship.